GNBF5020

Complete Revision Pack

Introduction to Molecular Biology and Genetics
Lectures 1-5 | 12 Sep - 17 Oct 2026

5 lectures. Audio narrative + slide content + concept-map diagrams + cheat sheet + active-recall flashcards. All in one printable PDF, redesigned for fast reading and easy recall.

Audio source: per-lecture MP4 -> ASR transcripts
Slide source: 454 frame-stills (1/90s) -> vision-Q&A; -> deduplicated
Diagrams: 16 full-page plates - 12 concept maps + 4 pathway maps
Cards: this PDF keeps each concept <= 2 short sentences

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>> 1. Figure plates (one full page each, printed 1:1)

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>> 6. Lecture 5 - Control of Gene Expression

>> 7. Cross-lecture Cheat Sheet

>> 8. Case Study Bank (16 worked scenarios)

>> 9. Active-Recall Flashcards (120 Q/A)

Figures in this pack:

- Course Map
- Lecture 1 - Central Dogma & Biomolecule Classes
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- Lecture 5 - Three Environmental Sensors

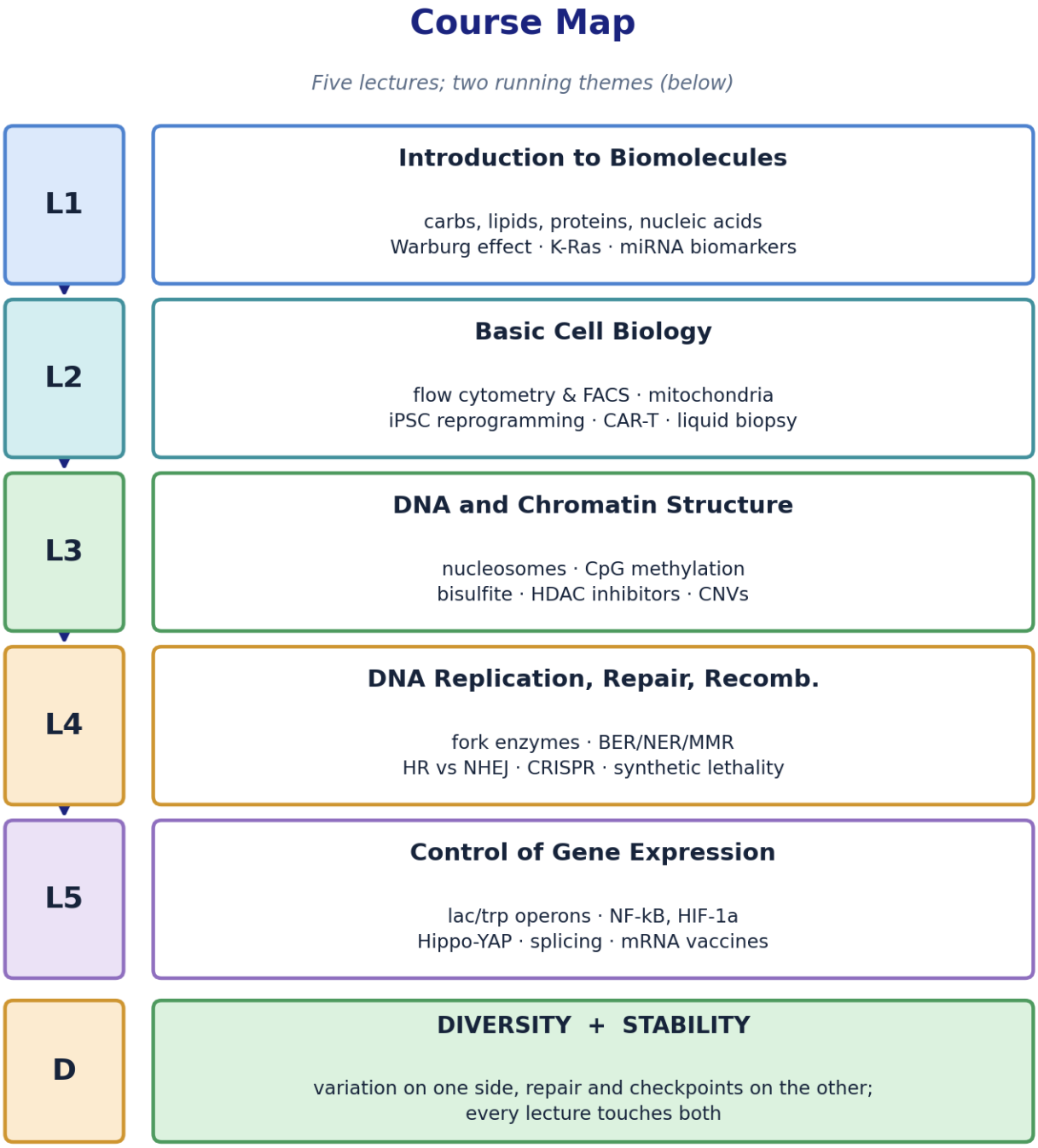
Video guides

Sixteen terms in this book carry the most exam weight, and each one has a short video guide: **English narration, Traditional Chinese subtitles**, about a minute each. The MP4 files live in the project folder **_video/assets/** - the full path is printed next to every link below, so you can still find a file after this PDF has been moved. **_video/index.html** plays all Sixteen on one page.

Video	Term	Chapter	Video file
V01	**Kaplan–Meier curve**	Ch 10 · Cancer biology and targeted therapy	_video/assets/V01_Kaplan_Meier_curve.mp4
V02	**Contractile ring → binucleated cell**	Ch 8 · Cell biology	_video/assets/V02_Contractile_ring_binucleated_cell.mp4
V03	**NHEJ defect → SCID**	Ch 6 · DNA damage and repair	_video/assets/V03_NHEJ_defect_SCID.mp4
V04	**Mitochondrial pedigree**	Ch 8 · Cell biology	_video/assets/V04_Mitochondrial_pedigree.mp4
V05	**Meiosis I vs Meiosis II (M1 vs M2)**	Ch 8 · Cell biology	_video/assets/V05_Meiosis_I_vs_Meiosis_II_M1_vs_M2.mp4
V06	**BER vs NER**	Ch 6 · DNA damage and repair	_video/assets/V06_BER_vs_NER.mp4
V07	**Hot tumour vs cold tumour**	Ch 9 · Immunology and cancer immunity	_video/assets/V07_Hot_tumour_vs_cold_tumour.mp4
V08	**Driver vs passenger**	Ch 10 · Cancer biology and targeted therapy	_video/assets/V08_Driver_vs_passenger.mp4
V09	**IgG1 vs IgG4**	Ch 9 · Immunology and cancer immunity	_video/assets/V09_IgG1_vs_IgG4.mp4
V10	**Predictive vs prognostic biomarker**	Ch 11 · Assay techniques and omics	_video/assets/V10_Predictive_vs_prognostic_biomarker.mp4
V11	**cis vs trans**	Ch 7 · Control of gene expression	_video/assets/V11_cis_vs_trans.mp4
V12	**de novo vs maintenance methylation**	Ch 4 · Chromatin and epigenetics	_video/assets/V12_de_novo_vs_maintenance_methylation.mp4
V13	**DNA replication**	Ch 5 · DNA replication	_video/assets/V13_DNA_replication.mp4
V14	**Transcription**	Ch 7 · Control of gene expression	_video/assets/V14_Transcription.mp4
V15	**Translation**	Ch 2 · Proteins and enzymes	_video/assets/V15_Translation.mp4
V16	**DNA repair**	Ch 6 · DNA damage and repair	_video/assets/V16_DNA_repair.mp4

Open the video index page ->

Course Map



All five lectures at a glance, framed by the two running themes.

Lecture 1 - Central Dogma & Biomolecule Classes

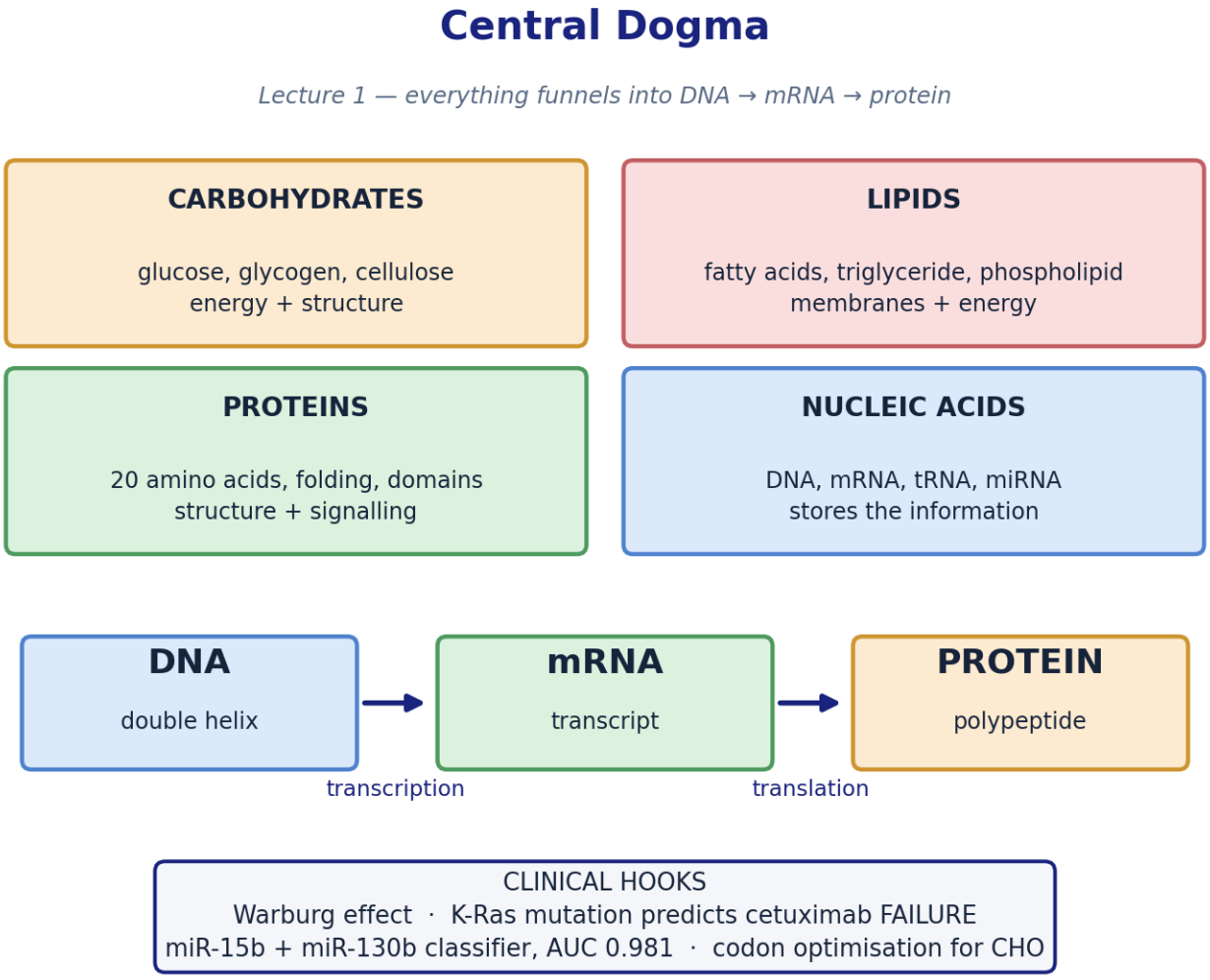


Figure 1.1 - Everything in Lecture 1 funnels into DNA -> mRNA -> protein.

Lecture 1 - The Warburg Effect

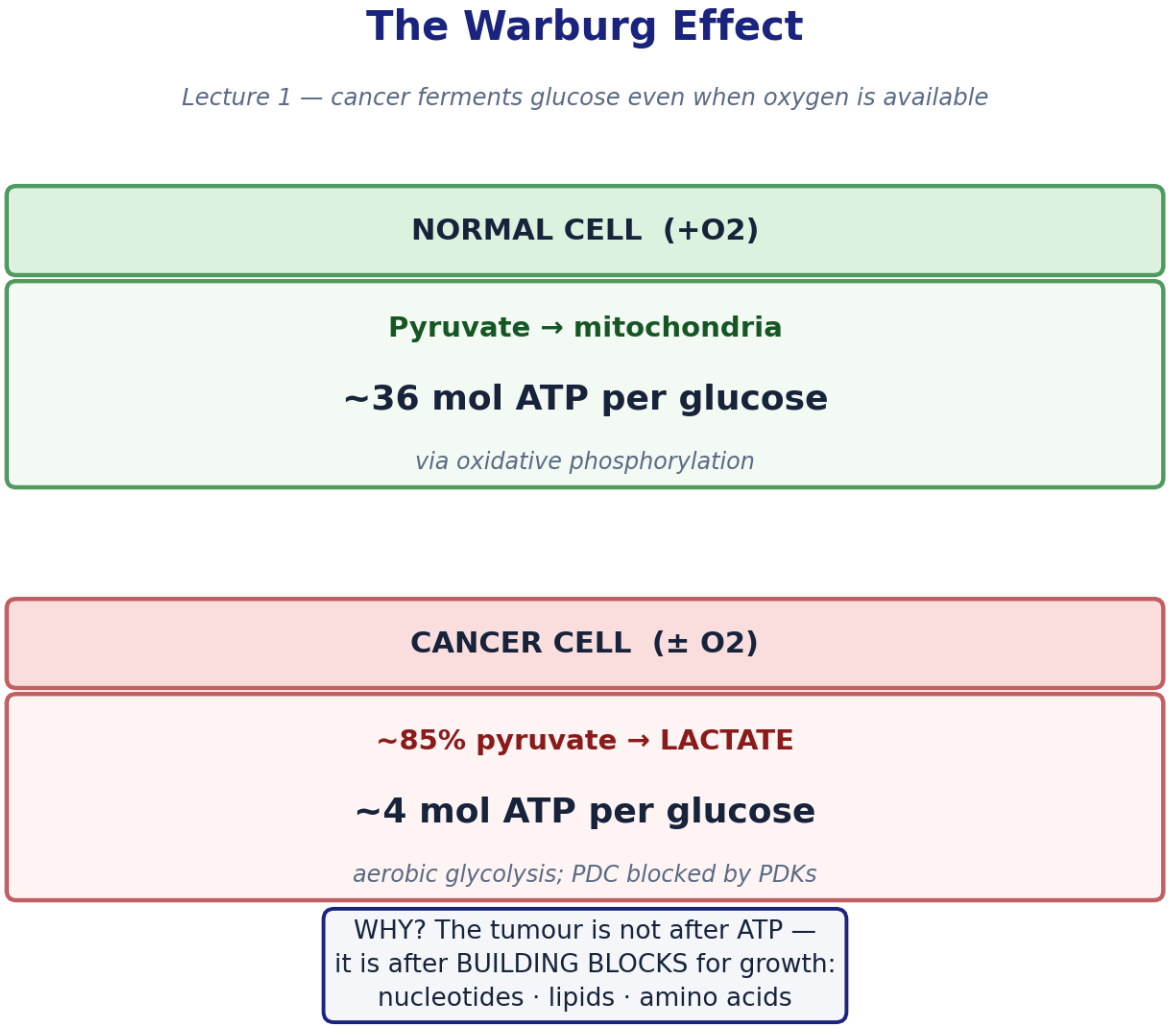


Figure 1.2 - Cancer cells ferment glucose to lactate even with oxygen present.

Lecture 1 - Hippo-YAP & K-Ras

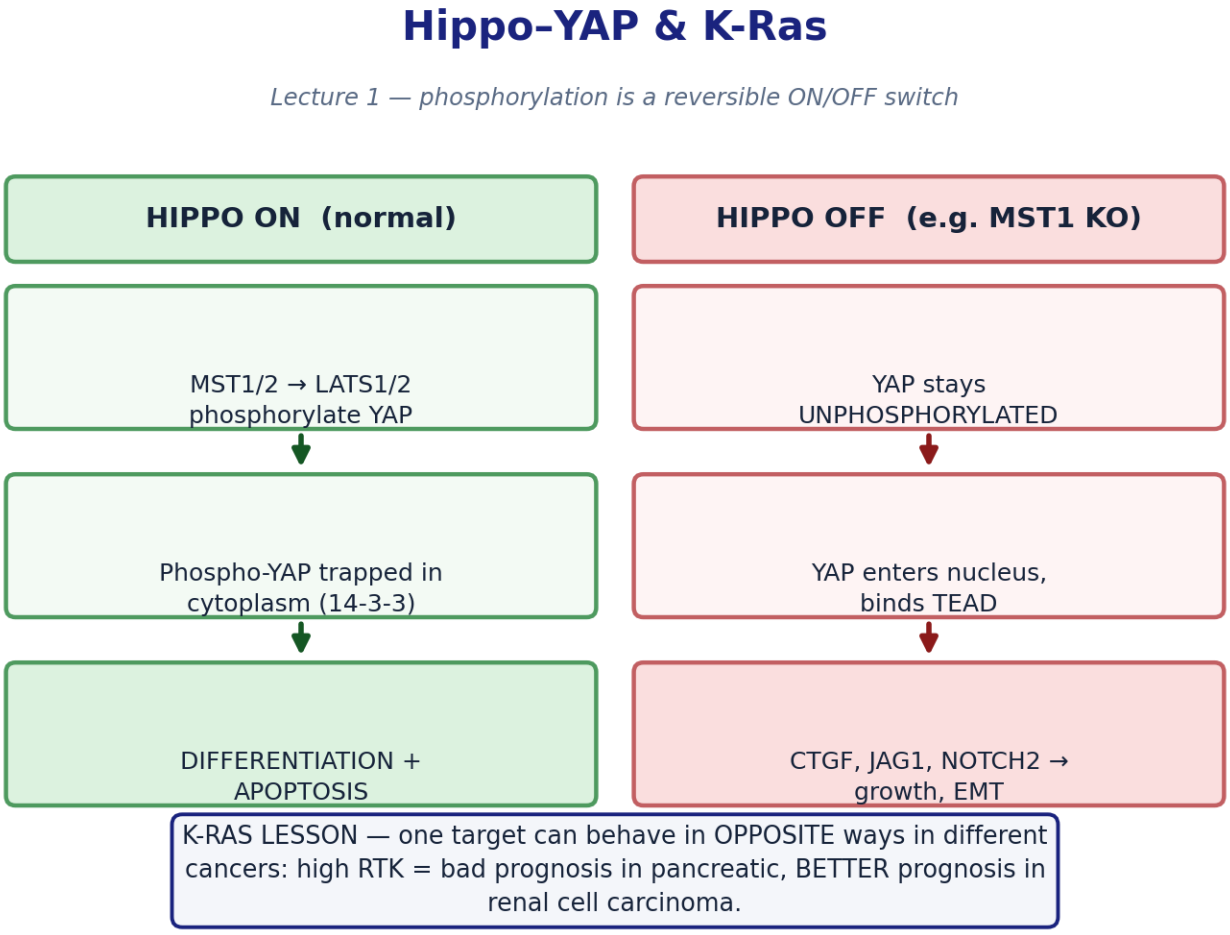


Figure 1.3 - Phosphorylation as a reversible switch; why a target can behave in opposite ways in different cancers.

Lecture 2 - Cell Cycle & Measurement Tools

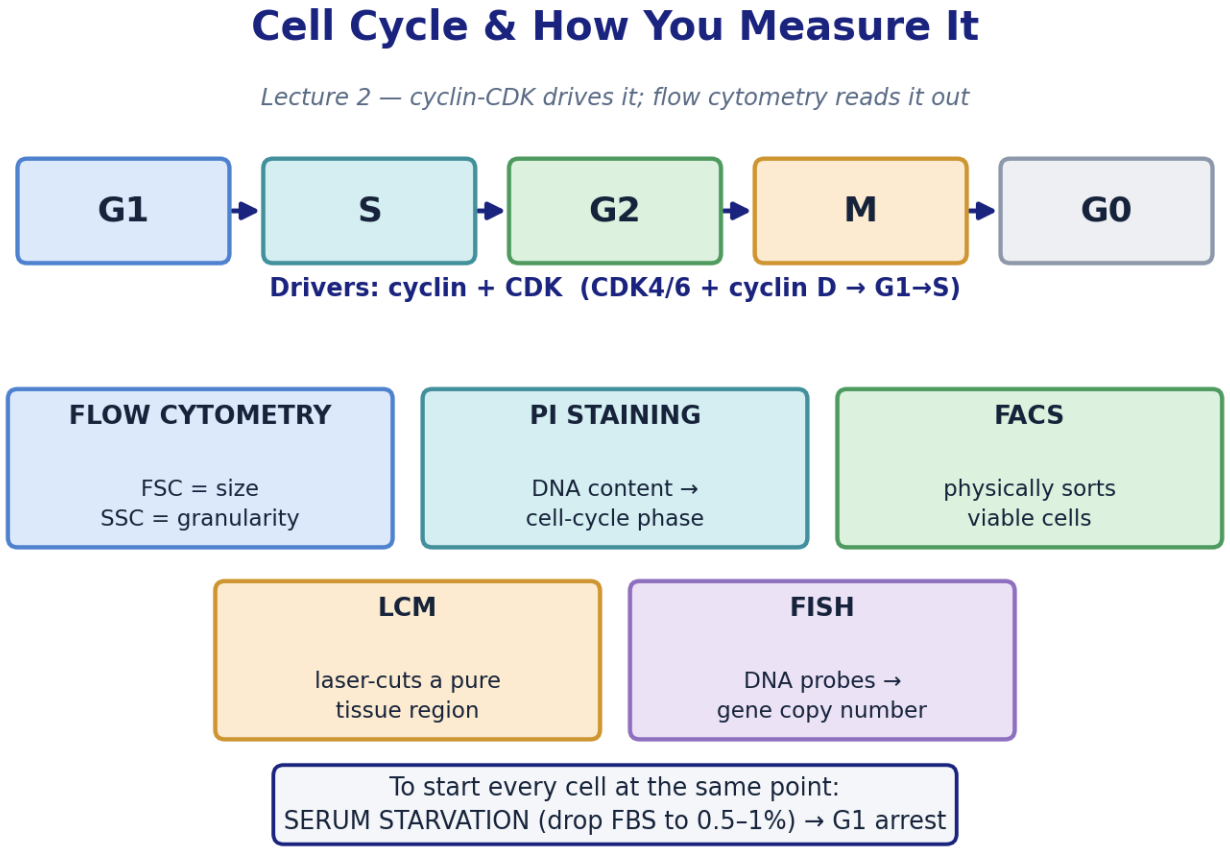


Figure 2.1 - Cyclin-CDK drives the cycle; flow cytometry and PI read it out.

Lecture 2 - Mitochondria & iPSC Reprogramming

Two Stories: Mitochondria & iPSC

Lecture 2 — one explains inherited disease, the other builds cell types

MITOCHONDRIA

Endosymbiotic origin — once a free bacterium

Maternal inheritance ONLY; paternal mtDNA is destroyed after fertilisation

"Mitochondrial Eve" = one maternal ancestor

THRESHOLD EFFECT: disease appears only when the mutant:normal ratio exceeds tissue tolerance

iPSC REPROGRAMMING (Yamanaka)

Somatic cell (e.g. fibroblast) + 4 factors

Oct3/4 · Sox2 · Klf4 · c-Myc (= OSKM)

GOLD-STANDARD TEST: teratoma assay in a mouse → tissues from all 3 germ layers

Uses: disease modelling, cardiotoxicity screening, tissue engineering

Downstream: engineered CAR-T (scFv + CD3z) · CRISPR ex-vivo gene therapy · liquid biopsy (EpCAM+/CD45- CTCs — watch EMT)

Figure 2.2 - Inherited disease via the threshold effect; Yamanaka OSKM factors.

Lecture 3 - DNA Packaging Hierarchy

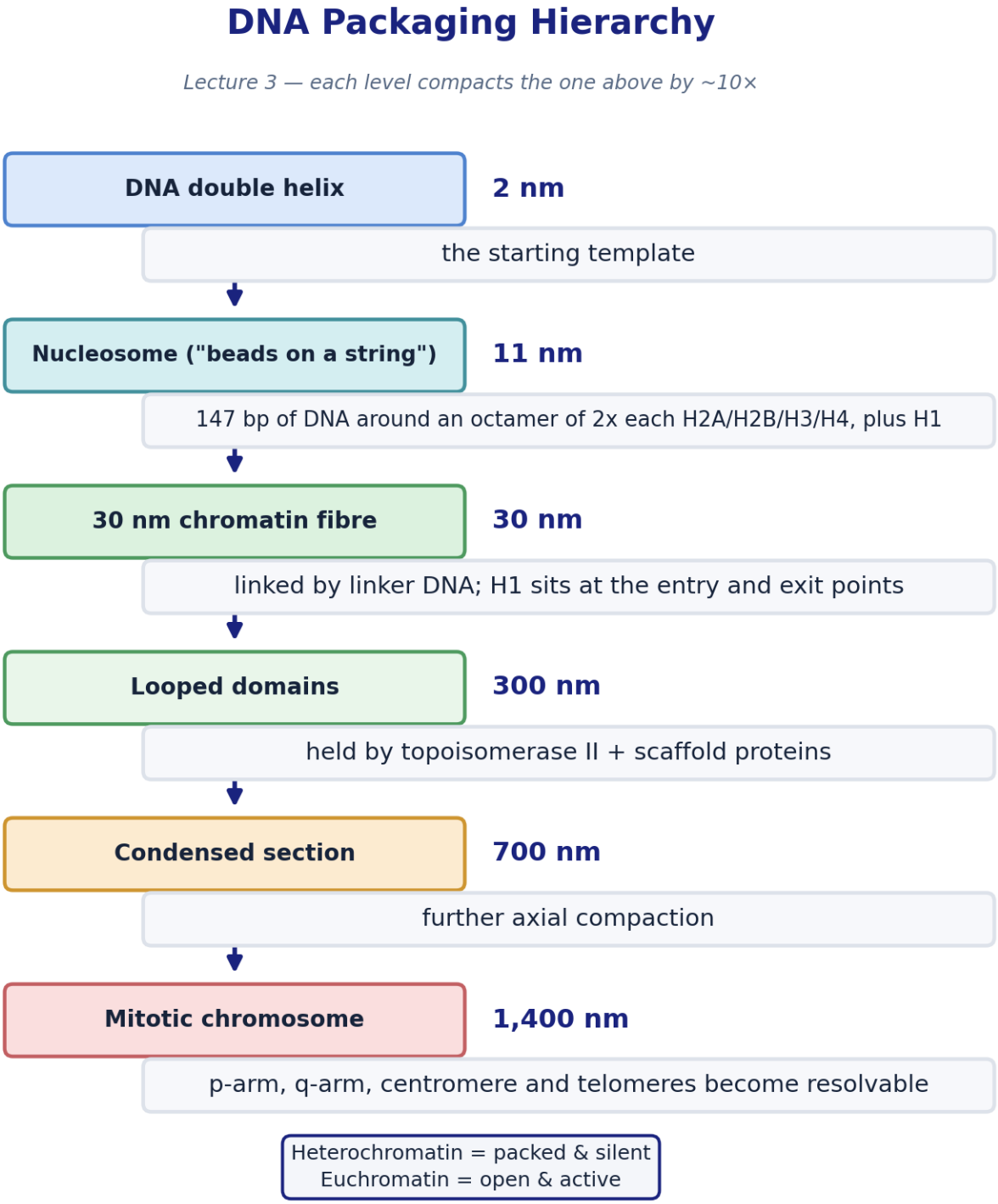


Figure 3.1 - From the 2 nm helix to the 1,400 nm mitotic chromosome.

Lecture 3 - Epigenetic Marks as Biomarkers & Drugs

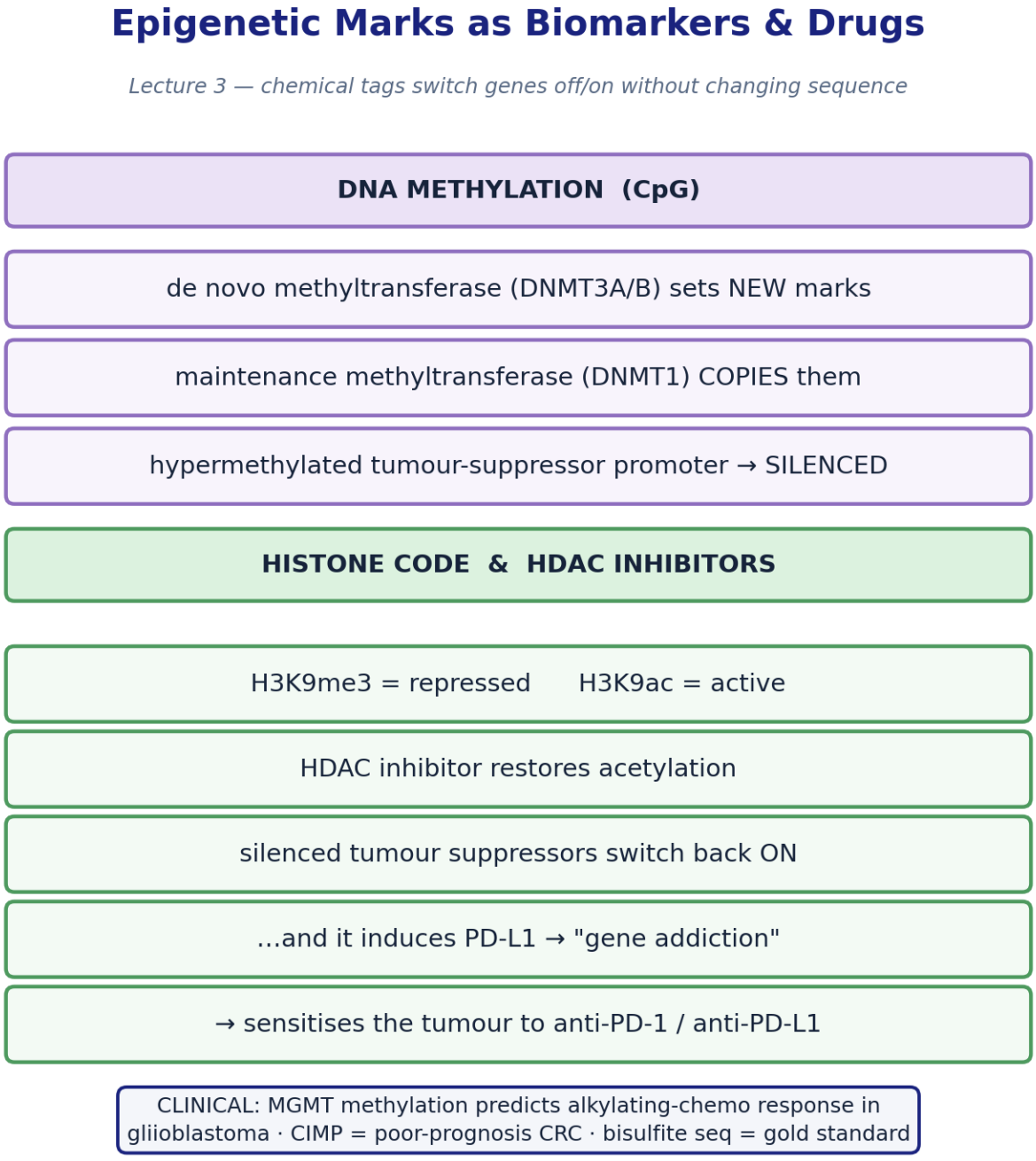


Figure 3.2 - CpG methylation, the histone code, and HDAC inhibitors.

Lecture 4 - The Replication Fork

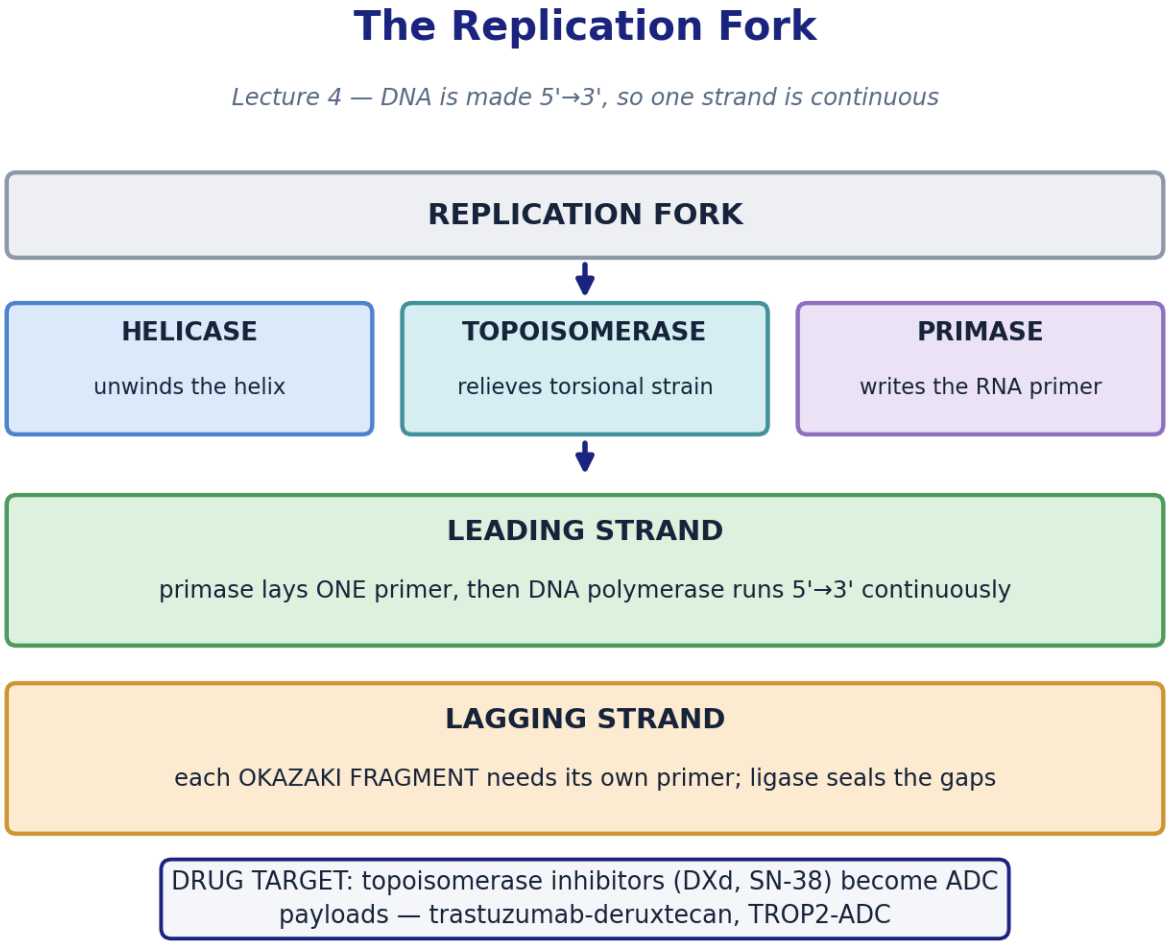


Figure 4.1 - Leading vs lagging strand; helicase, topoisomerase, primase.

Lecture 4 - Which DNA Repair Pathway?

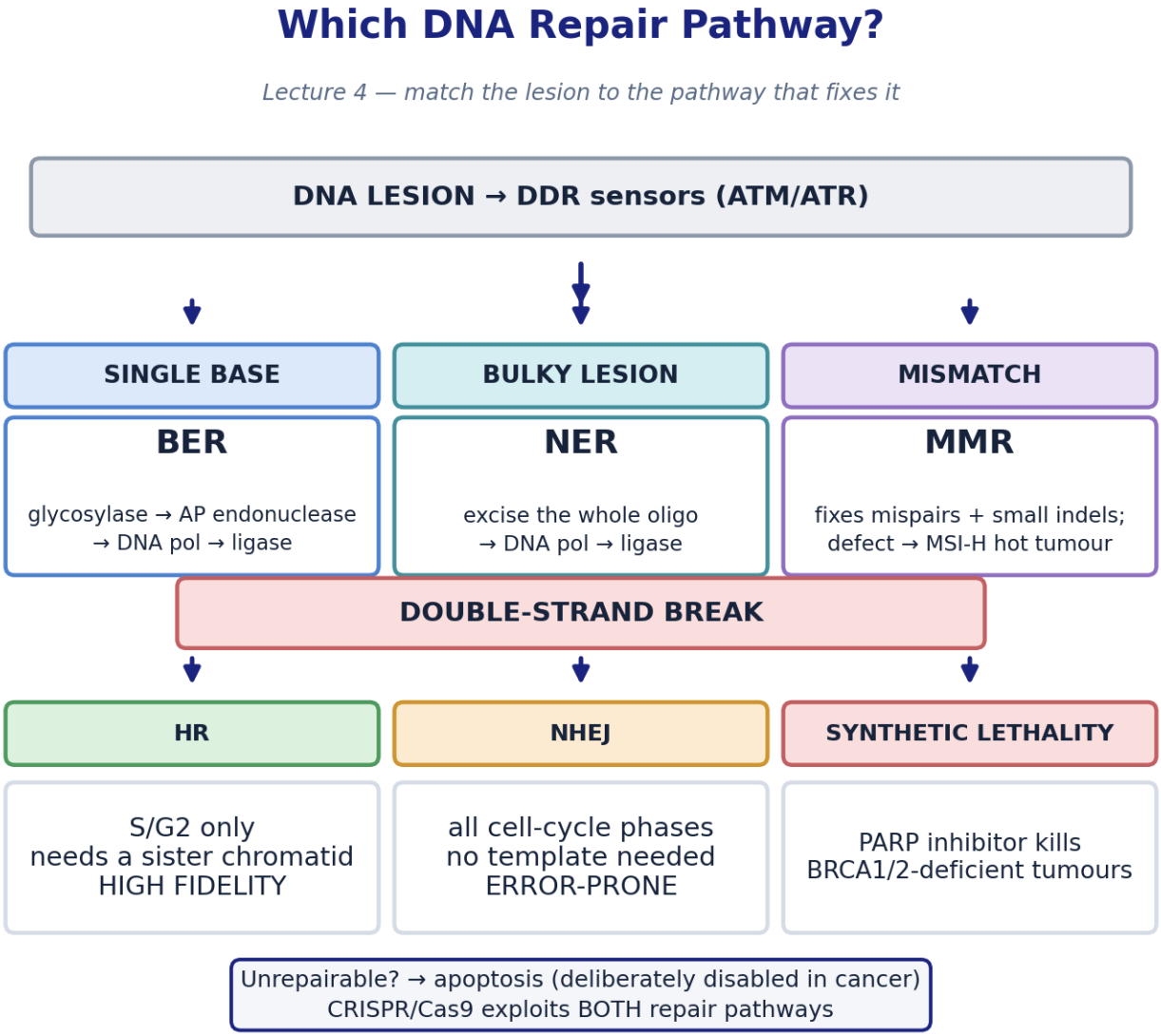


Figure 4.2 - Match the lesion to BER / NER / MMR, then HR vs NHEJ.

Lecture 5 - Lac Operon: Four States

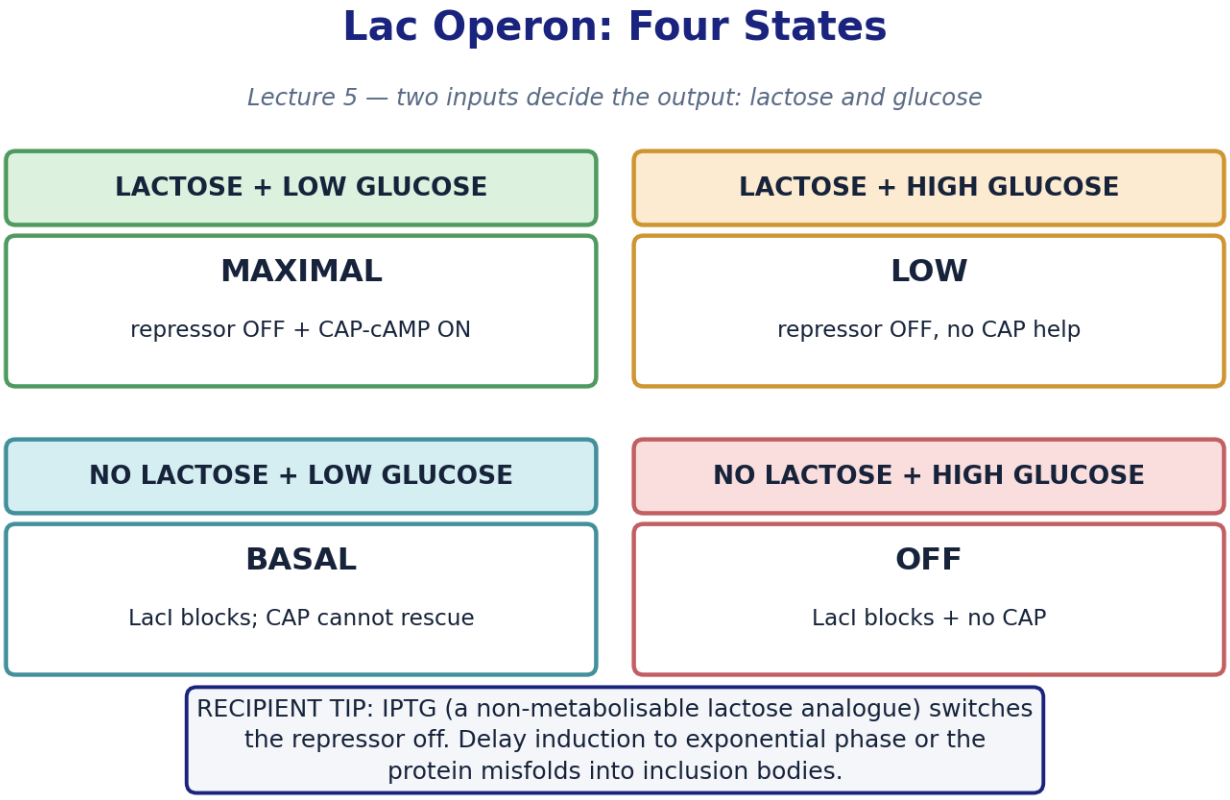


Figure 5.1 - Lactose and glucose together decide the output.

Lecture 5 - Three Environmental Sensors

Three Environmental Sensors Become TFs

Lecture 5 — each pathway reads the outside world and rewires genes

NF-KAPPA-B (inflammation)

Held in the cytoplasm by I-KAPPA-B

LPS / TNF signal → I-KAPPA-B degraded

NF-KAPPA-B enters the nucleus

Cytokines: TNF-alpha → IL-1 beta → IFN-gamma → IL-2 → IL-15

HIF-1-ALPHA (hypoxia)

Normoxia: destroyed immediately

Hypoxia: stabilised → enters nucleus

Binds HREs → induces VEGF

TRICK: HRE-controlled CAR-T fires only in the hypoxic tumour

HIPPO – YAP (cell density)

Dense sheet → MST1 → LATS1 phosphorylate YAP

Phospho-YAP stays in the cytoplasm

…so no proliferation

Pathway lost → YAP in nucleus → CTGF / JAG1 / NOTCH2 → overgrowth

ENGINEERED: synNotch AND-gate CAR-T · personalised mRNA vaccines · CRISPR ex-vivo gene editing

Figure 5.2 - NF-kB, HIF-1a and Hippo-YAP all rewire gene expression.

Pathway 1 - Receptor tyrosine kinase signalling (EGFR / RAS / PI3K)

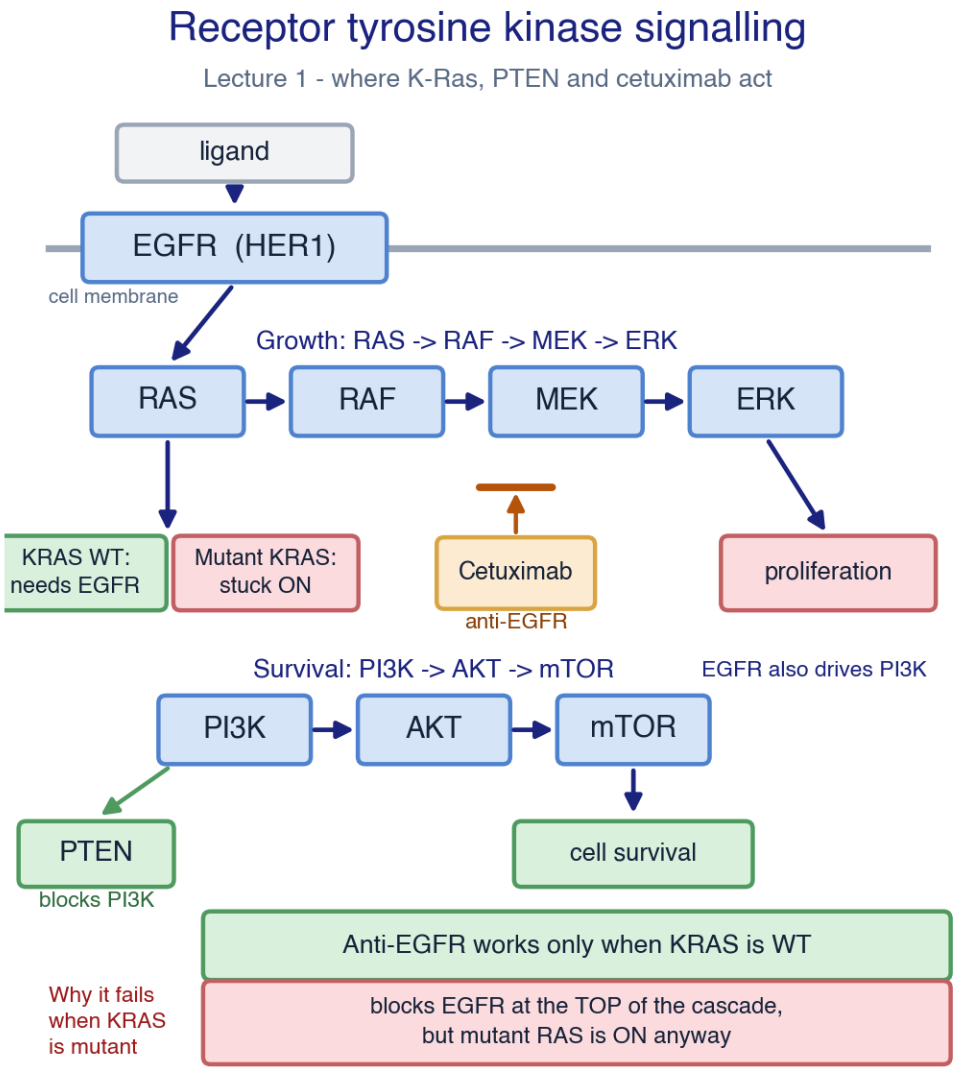


Figure P1 - Lecture 1. The cascade that K-Ras, PTEN and cetuximab all act on.

Pathway 2 - LPS / TLR4 to NF-kB

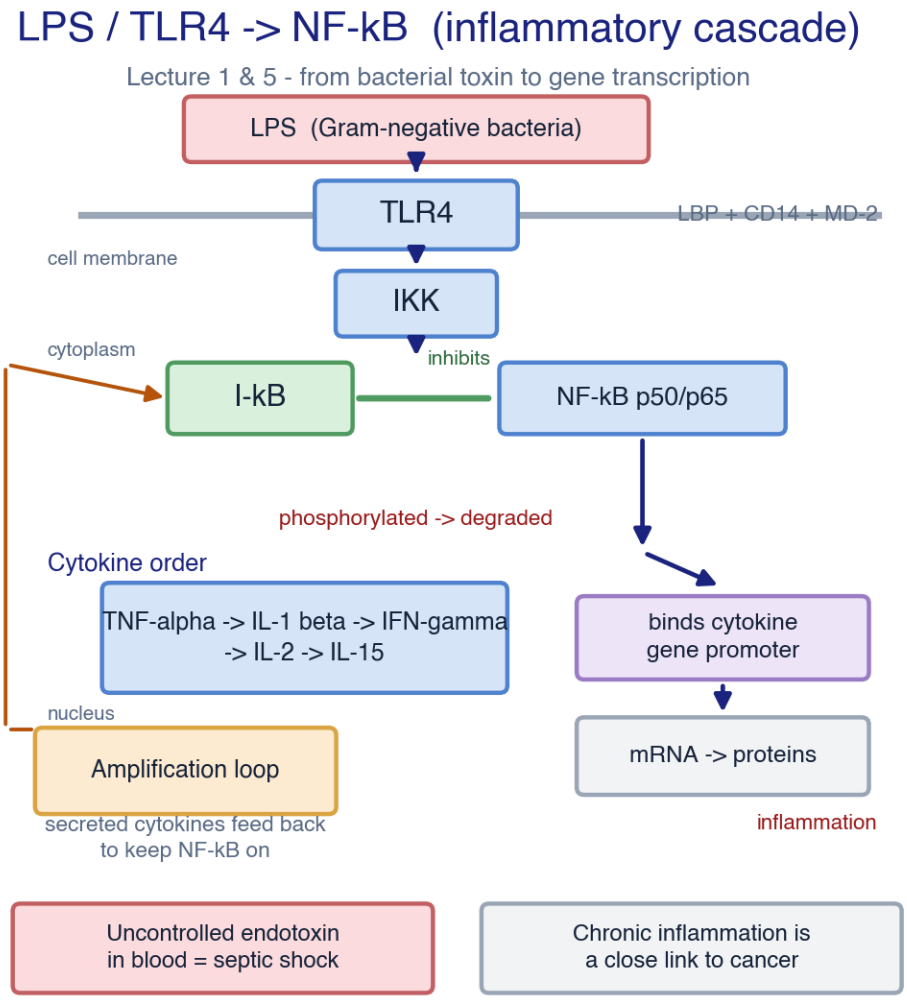


Figure P2 - Lectures 1 and 5. From bacterial toxin to inflammatory gene transcription.

Pathway 3 - Nuclear receptor signalling (Type I vs Type II)

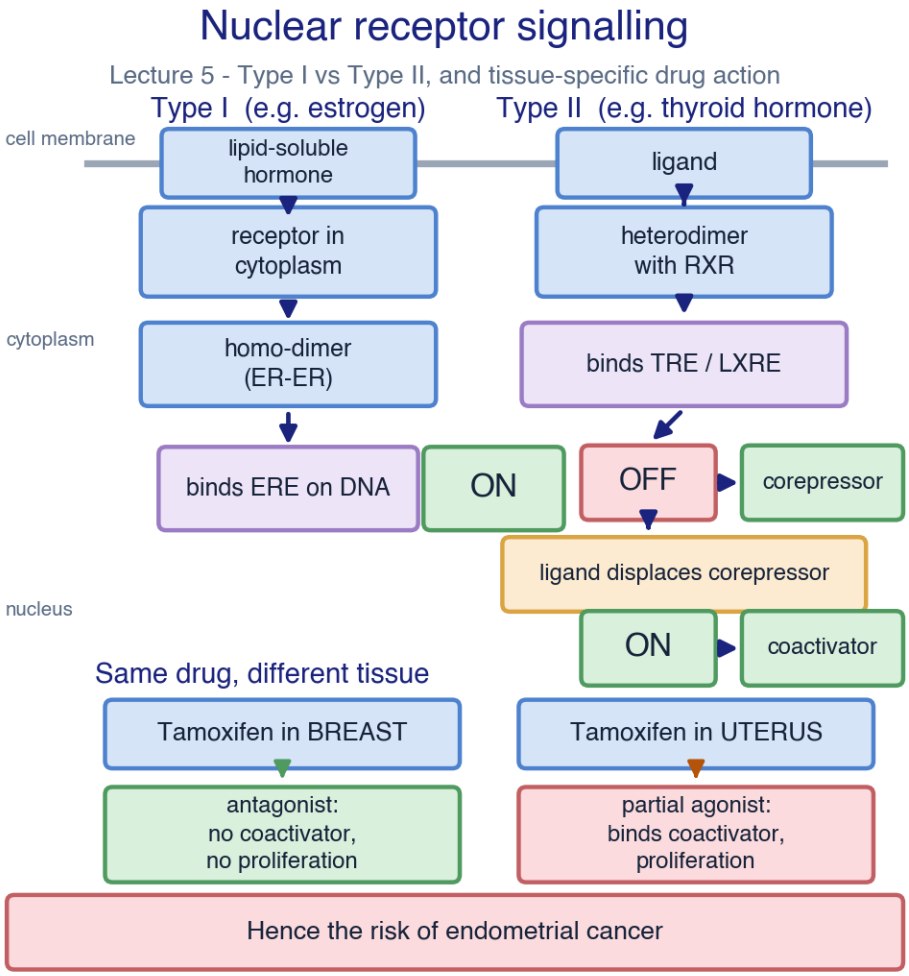


Figure P3 - Lecture 5. Why tamoxifen acts differently in breast and uterine tissue.

Pathway 4 - Hypoxia sensing by HIF-1a

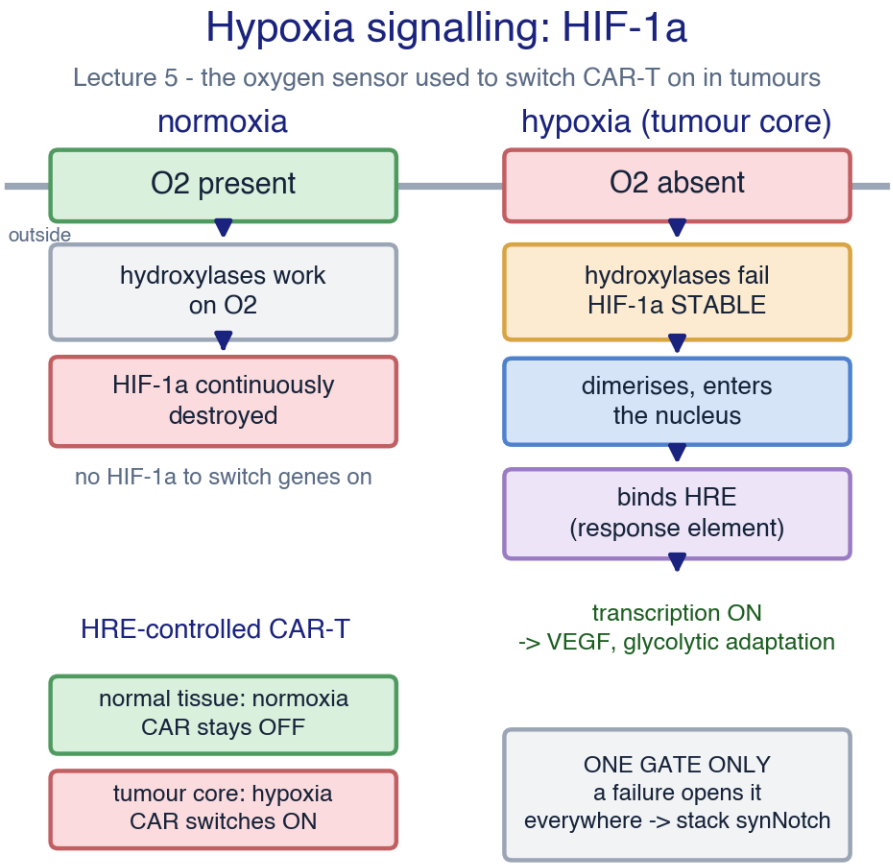


Figure P4 - Lecture 5. The oxygen sensor used to switch CAR-T on inside a tumour.

1. Lecture 1: Introduction to Biomolecules

Date: 12 Sep 2026 | **Duration:** ~2h 24m | **Instructor:** Dr Tony Wong

THEME: Carbs, lipids, proteins, nucleic acids; everything connects to clinical biomarkers and the central dogma.

What this lecture covers

- Lecture 1 - Central Dogma & Biomolecule Classes
- Lecture 1 - The Warburg Effect
- Lecture 1 - Hippo-YAP & K-Ras

> **[INFO]** Figures for this lecture have their own full pages in the plate section, printed at 1:1 so the labels are at full size. Slide-specific facts (numbers, citations) are woven into the concept cards and the If-you-see/think-that table.

Lecture 1 - Core Concepts

Warburg effect

Cancer cells ferment glucose to lactate even with O2 present. Frees intermediates for biosynthesis. **~4 mol ATP** vs **~36 mol** in normal oxidative phosphorylation.

Carbohydrates

Glucose (C6H12O6, pyranose), fructose (furanose). Glycogen has alpha-1,4 linkages with alpha-1,6 branches. Cellulose = structural.

Lipids / LPS-TLR4-NFkB

Saturated vs unsaturated FAs -> triglycerides or phospholipid bilayer. LPS of Gram-negatives binds TLR4 -> NFkB -> septic shock.

Protein hierarchy

Primary (sequence) -> Secondary (alpha-helix/beta-sheet) -> Tertiary (3D fold) -> Quaternary (subunits). Domain = independent fold. Methods: Edman (legacy) / LC-MS/MS.

Antibody isotypes

IgG1 -> ADCC (Fc engages NK). IgG4 -> no ADCC, longer half-life. ADCs = antibody + linker + cytotoxic payload (e.g. trastuzumab-deruxtecan).

Hippo-YAP

MST1 -> LATS1 phosphorylate YAP -> YAP cytoplasm = no proliferation. MST1 KO -> YAP nucleus -> overgrowth (Lu L et al., 2010).

Central dogma

DNA -> mRNA -> protein. Codon optimisation replaces synonymous codons for host (e.g. CHO).

K-Ras + miRNA

Mutant K-Ras = constitutively active -> **cetuximab (anti-EGFR) fails**. New drug 'Rasonque' (mock) hit OS 13.2 vs 6.7 mo. miRNAs stable in serum -> non-invasive biomarkers.

[EXAM WATCH] Same target can mean opposite things in different cancers. RTK high = bad in pancreatic, *better* in renal cell carcinoma.

[CLINICAL] Avery-Macleod-McCarty (1943): DNase destroyed the transforming principle; proteases did not -> DNA is the hereditary material.

[EXAM WATCH] miR-15b + miR-130b classifier: **AUC = 0.981** (sensitivity 98.25%, specificity 91.53%) for HCC detection from serum. Liu AM et al., 2012.

Lecture 1 - Slide-Specific Facts & 'If you see -> think'

Same target, opposite prognosis	RTK: bad in pancreatic, good in renal cell.
Mutant K-Ras CRC + cetuximab	Don't give it - downstream K-Ras still on.
Tumor with TMN stage + miR-15b/130b high	Cutoffs: p=0.0158 (miR-130b), p=0.0637 (miR-15b).
Mutant KRAS + G12C variant	~13% of lung adenocarcinoma, ~3% of CRC.
Vimentin in HCC serum (Sun S 2010)	Controls 176, HCC<=2cm 271.5, HCC>2cm 231.3 ng/mL.
Slide 28 Hippo diagram (Patel SH 2017)	MST1/2 -> LATS1/2 -> P-YAP -> 14-3-3 sequestration; Hippo OFF -> YAP+TEAD -> CTGF/JAG1/NOTCH2.

Lecture 1 - Quick Self-Test

Cover the answer column. Say it out loud. Then reveal.

Q1. Which base pair has 3 hydrogen bonds? A-T or G-C?	G-C. (A-T has 2.) Higher GC = higher melting temp T(m) - matters for PCR primer design.
Q2. What is codon optimisation and why is it needed in CHO cells?	Replace synonymous codons with host-preferred ones. Human proteins in CHO have different codon-usage bias - optimisation lifts yield.
Q3. Mutant K-Ras patient - do you give cetuximab?	No. Cetuximab blocks EGFR; mutant K-Ras is constitutively active downstream of EGFR. Use a K-Ras inhibitor instead.
Q4. What does the Warburg effect give a cancer cell?	~4 mol ATP/glucose, plus free intermediates (nucleotides, lipids, AAs) for biosynthesis. Lactate is the biomarker in metabolomics.
Q5. Name the Hippo cascade and what happens to YAP.	MST1 -> LATS1 phosphorylate YAP. Phospho-YAP is sequestered in cytoplasm (14-3-3) or degraded. Loss of MST1 -> YAP enters nucleus -> overgrowth/cancer.

2. Lecture 2: Basic Cell Biology

Date: 19 Sep 2026 | **Duration:** ~2h 20m | **Instructor:** Dr Tony Wong

THEME: Tools to isolate cell populations; mitochondria and iPSC biology; engineered CAR-T and liquid biopsy.

What this lecture covers

- Lecture 2 - Cell Cycle & Measurement Tools
- Lecture 2 - Mitochondria & iPSC Reprogramming

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Lecture 2 - Core Concepts

Flow cytometry

FSC = cell size; SSC = granularity. PI = DNA content. Gating isolates subsets (e.g. CD3+CD4+ helper T cells).

FACS / LCM / FISH

FACS physically sorts viable cells. LCM = laser-captured tissue region. FISH = DNA probes for gene copy number (HER2 amplification test).

Mitochondria

Endosymbiotic origin. **Maternal inheritance only** (paternal mtDNA destroyed post-fertilisation). Threshold effect in disease.

Cell cycle

G1 -> S -> G2 -> M -> cytokinesis. Drivers: **cyclin-CDK**. Synchronise via **serum starvation** (0.5-1% FBS). PI staining measures phase.

Meiosis vs mitosis

Mitosis = identical daughter cells. Meiosis I = homologues, with **crossing over** -> diversity. Meiosis II = sister chromatids.

Yamanaka iPSC

OSKM (Oct3/4, Sox2, Klf4, c-Myc) reprogram somatic cells -> iPSC. Validate via **teratoma** (3 germ layers in mice).

CRISPR vs siRNA

CRISPR = permanent (DSB + repair). siRNA/shRNA = transient (RISC knockdown).

CAR-T flavours

Autologous (patient), **allogeneic** (off-the-shelf), **in vivo** (mRNA-LNP). scFv + intracellular CD3z/4-1BB/CD28.

Liquid biopsy

CTCs: **EpCAM+ / CD45-** selection. Watch for **EMT** changing markers - CTC profile can differ from primary tumour.

[CLINICAL] MSI-high tumours (deficient MMR) are 'hot' - many frameshift neoantigens - respond to anti-PD-1/PD-L1 checkpoint blockade.

[EXAM WATCH] Divide cytoplasm failure (cytokinesis blocked by inhibitor) -> binucleated cells. Distinct from nuclear division failure.

[CLINICAL] Drug development: target nomination needs disease-specific overexpression AND structural tractability (small molecule / antibody / ASO).

Lecture 2 - Slide-Specific Facts & 'If you see -> think'

FSC / SSC in flow cytometry	Size / granularity.
Bruton / SCID patient, gene defect	NHEJ defect -> no V(D)J -> no mature T or B cells.
Binucleated cell after drug treatment	Cytokinesis blocked (contractile ring inhibited).
Extract cell populations from heterogeneous tumor	FACS / LCM / gradient - all standard.
Mitochondrial disease with low heteroplasmy	Below tissue threshold -> asymptomatic.
Pseudohypoxic HIF-1alpha stable tumor core	HIF -> HREs -> VEGF (anti-VEGF bevacizumab).

Lecture 2 - Quick Self-Test

Cover the answer column. Say it out loud. Then reveal.

Q1. FSC vs SSC in flow cytometry?	FSC = cell size; SSC = internal complexity/granularity.
Q2. Why is mtDNA inherited only from the mother?	Paternal mitochondria are tagged (ubiquitin) and destroyed after fertilisation.
Q3. What's the gold-standard in vivo test for iPSC pluripotency?	Teratoma assay: inject into immunodeficient mouse -> 3 germ-layer tissues.
Q4. How do you isolate circulating tumour cells (CTCs)?	EpCAM+ positive selection (epithelial tumour), CD45- negative selection (exclude WBCs).
Q5. Why does mutant KRAS predict cetuximab failure?	Mutant K-Ras is constitutively active downstream of EGFR - blocking the receptor upstream does not stop the signal.

3. Lecture 3: DNA and Chromatin Structure

Date: 3 Oct 2026 | **Duration:** ~2h 6m | **Instructor:** Dr Tony Wong

THEME: From nucleosome to chromosome; epigenetic marks as biomarkers and therapeutic targets.

What this lecture covers

- Lecture 3 - DNA Packaging Hierarchy
- Lecture 3 - Epigenetic Marks as Biomarkers & Drugs

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Lecture 3 - Core Concepts

Base pairing

G-C = **3 H-bonds**; A-T = 2. Higher GC = higher T(m). Critical for **PCR primer design**.

Packaging hierarchy

DNA helix 2 nm -> nucleosome 11 nm -> 30 nm fibre -> looped 300 nm -> 700 nm -> metaphase chromosome 1,400 nm.

Nucleosome

147 bp DNA wrapped around H2A/H2B/H3/H4 octamer (2 of each) + H1 linker. Luger et al. 1997 crystal at **2.8 Angstroms**.

Histone code (naming)

H3K9me3 = Histone 3, Lysine-9, tri-methylation = repressed. H3K9ac = active.

DNA methylation

De novo methyltransferases (DNMT3A/B) set new marks; maintenance (DNMT1) copies to daughter strands.

MGMT promoter methylation

Silences MGMT DNA-repair enzyme. Glioblastoma becomes sensitive to **alkylating chemotherapy + RT**.

HDAC inhibitors

Restore acetylation -> re-express silenced tumour suppressors. Induces PD-L1 -> 'gene addiction' -> sensitive to anti-PD-1/PD-L1.

Chromosome anatomy

p-arm (short), q-arm (long), **centromere** (microtubule attachment), **telomere** (TTAGGG x 3-20 kb).

G-band nomenclature

q21.1 = 'q-two-one-point-one', NOT 'q-twenty-one-point-one'.

Telomere / Hayflick

Shortens each division -> Hayflick limit -> senescence. Cancer reactivates **telomerase** (reverse transcriptase with internal RNA template).

[EXAM WATCH] CIMP (CpG Island Methylator Phenotype) = poor prognosis CRC. Bisulfite sequencing = gold-standard methylation profiling.

[CLINICAL] CNV example: AMY1 amylase - high copy number in high-starch-eating populations, generated by **NAHR** (non-allelic homologous recombination).

[EXAM WATCH] Distinguish driver (causal) vs passenger (co-amplified) genes by integrating CNV profile + mRNA expression. NPM1 in leukaemia is a textbook example.

Lecture 3 - Slide-Specific Facts & 'If you see -> think'

G-C content	3 H-bonds; higher T(m); harder to denature.
q21.1 read-out	q-two-one-point-one - NOT q-twenty-one-point-one.
Histone mark H3K9me3	Repressed.
Chemo-resistant if hypermethylated MGMT?	No - hypermethylated MGMT actually SENSITISES to alkylating agents.
CIMP in colorectal cancer	CpG island hypermethylation across tumour-suppressor promoters -> poor prognosis.
CNV generation in meiosis	NAHR (non-allelic homologous recombination) - unequal crossover at repeats.
iPSC validation test	Teratoma assay in mice - 3 germ layers.

Lecture 3 - Quick Self-Test

Cover the answer column. Say it out loud. Then reveal.

Q1. How many H-bonds in G-C vs A-T?	G-C = 3, A-T = 2. Higher GC content = higher melting temperature.
Q2. What's a 147-bp particle wrapping an octamer of 2x(H2A/H2B/H3/H4) + 1 H1 linker?	A nucleosome.
Q3. Name the methylation writer for new marks vs the one for copying marks to daughters.	De novo: DNMT3A/3B. Maintenance: DNMT1.
Q4. A glioblastoma patient has MGMT promoter methylation. What does it predict?	Sensitivity to alkylating chemotherapy + RT - better survival.
Q5. How do you distinguish driver from passenger CNVs in cancer?	Integrate CNV with mRNA expression - true drivers show concordant copy gain + transcript upregulation (e.g. NPM1 in leukaemia).

4. Lecture 4: DNA Replication, Repair and Recombination

Date: 10 Oct 2026 | **Duration:** ~2h 18m | **Instructor:** Dr Tony Wong

THEME: Replicate faithfully; repair damage; exploit repair biology for therapy.

What this lecture covers

- Lecture 4 - The Replication Fork
- Lecture 4 - Which DNA Repair Pathway?

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Lecture 4 - Core Concepts

Semi-conservative replication

Meselson-Stahl. Each daughter has 1 old + 1 new strand.

Core enzymes

Helicase (unwind) -> Topoisomerase (relieve torque) -> Primase (RNA primer) -> DNA Pol (5'->3' + 3'->5' proofreading) -> Ligase (seal nicks).

Leading / lagging

Leading = continuous 5'->3'. **Lagging** = discontinuous **Okazaki fragments** (each primed separately).

Telomere biology

TTAGGG x 3-20 kb at chromosome ends. Shortens each round. **Telomerase** = reverse transcriptase with internal RNA template.

Damage classification

Single base -> BER. Bulky (UV dimers) -> NER. Mismatch -> MMR. DSB -> HR / NHEJ. ATM/ATR sensors.

BER vs NER

BER = single damaged base (glycosylase -> AP endonuclease -> Pol -> ligase). NER = bulky multi-nucleotide lesion (excise oligo).

MMR / MSI-H

Mismatch repair deficiency -> **microsatellite instability** -> frameshift burden -> neoantigens -> 'hot tumour' -> checkpoint inhibitor candidate.

HR vs NHEJ

HR = sister-chromatid template; S/G2 only. NHEJ = no template; works all phases; error-prone; basis of V(D)J recombination.

PARP inhibitor

BRCA1/2-deficient (HR-defective) tumours are synthetically lethal when PARP is blocked - wide therapeutic window.

CRISPR/Cas9

Cas9 + guide RNA = targeted DSB. **NHEJ** = indels = knockout. **HR + donor** = precise knock-in (S/G2 only).

[EXAM WATCH] ADCs: trastuzumab-deruxtecan uses topoisomerase inhibitor (DXd) payload against HER2. Dual-payload ADCs combine two agents to bypass single-drug resistance.

[CLINICAL] MSI-H status testing uses PCR + immunohistochemistry. MMR genes: MLH1, MSH2, MSH6, PMS2. Lynch syndrome is the hereditary form.

[EXAM WATCH] Telomere length re-extension in HCC is an independent prognostic factor for higher recurrence and worse survival.

Lecture 4 - Slide-Specific Facts & 'If you see -> think'

UV pyrimidine dimers	NER (bulky lesion).
O6-methylguanine from alkylating chemo	Repaired by MGMT - if MGMT methylated -> more damage -> better response.
NHEJ error-prone in S phase	Works all phases, not S/G2 only. (HR is S/G2 only.)
MSH2/6, MLH1, PMS2 deficiency	MMR-defective, MSI-H, frameshift neoantigens.
CRISPR / Cas9 + no donor template	NHEJ -> small indel -> gene knockout.
PARP inhibitor in BRCA1/2 cancer	Synthetic lethality: tumour can't use HR to fix PARP-trapped SSBs.

Lecture 4 - Quick Self-Test

Cover the answer column. Say it out loud. Then reveal.

Q1. Meselson-Stahl proved what?	Semi-conservative replication - each daughter has one old + one new strand.
Q2. Why does lagging-strand synthesis need primase?	DNA polymerase cannot start de novo - it needs a free 3'-OH, which primase creates by laying an RNA primer.
Q3. BER vs NER in one line.	BER fixes single damaged bases; NER fixes bulky multi-nucleotide lesions like UV pyrimidine dimers.
Q4. MSI-H tumour - which therapy class?	Immune checkpoint inhibitors (anti-PD-1 / anti-PD-L1).
Q5. CRISPR/Cas9 knockout vs knock-in pathways?	Knockout via NHEJ = small indels. Knock-in via HR + donor template (S/G2 only).

5. Lecture 5: Control of Gene Expression

Date: 17 Oct 2026 | **Duration:** ~2h 11m | **Instructor:** Dr Tony Wong

THEME: Every layer of gene expression is a drug target or biomarker; the central logic scales from Lac/Trp to personalised mRNA vaccines.

What this lecture covers

- Lecture 5 - Lac Operon: Four States
- Lecture 5 - Three Environmental Sensors

> **[INFO]** Figures for this lecture have their own full pages in the plate section, printed at 1:1 so the labels are at full size. Slide-specific facts (numbers, citations) are woven into the concept cards and the If-you-see/think-that table.

Lecture 5 - Core Concepts

Cis vs trans

Cis = same DNA molecule (promoter, enhancer). Trans = diffusible TF from another locus.

Lac operon

Lactose -> allolactose -> LacI off. Low glucose -> cAMP -> CAP -> on. IPTG = non-metabolised inducer for recombinant expression.

Trp operon

Repressible. Tryptophan binds Trp repressor -> shut down biosynthesis.

Eukaryotic transcription

Pol II + general TFs at TATA box. Enhancers/silencers can sit thousands bp away - DNA-bending proteins loop them to promoter.

Nuclear receptor

Architecture: **NTD / DBD / LBD**. Dimerisation boosts DNA binding. Herceptin blocks HER2 heterodimerisation.

Tamoxifen (SERM)

Antagonist in breast (anti-tumour); partial agonist in endometrium (small cancer risk).

Receptor occupancy (RO)

% of receptors that must be bound for effect. HER2: ~**2%** in breast vs ~**10%** in gastric cancer.

NF-kB cascade

Held by IkB in cytoplasm. LPS/TNF -> IkB degraded -> NFkB nucleus. Order: TNF-alpha -> IL-1 beta -> IFN-gamma -> IL-2 -> IL-15. Positive-feedback loop.

HIF-1alpha

Normoxia: degraded. Hypoxia: stabilises -> nucleus -> HREs -> VEGF. **Synthetic switch**: HRE-controlled CAR-T in hypoxic tumour only.

Hippo-YAP (revisited)

MST1 -> LATS1 phosphorylate YAP. Loss -> YAP nucleus -> CTGF/JAG1/NOTCH2 -> overgrowth.

Post-transcriptional

5' cap, poly-A tail length, 3' UTR. miRNAs bind 3' UTR seed -> mRNA decay / translational repression. Stable in serum = great biomarkers.

Modern applications

CAR-NK / CAR-M for solid tumours. **synNotch AND-gate** CAR-T (2 antigens = fire). **mRNA vaccines** (personalised by single-cell TCR seq).

[EXAM WATCH] The same SNP can create a novel splice site - HCC splice-variant case where shorter isoform -> worse survival. Always: alternative splicing = cancer-specific neoantigens + drug resistance.

[CLINICAL] Predictive biomarker = predicts response to therapy (e.g. K-Ras WT for cetuximab). Prognostic = predicts disease outcome independent of therapy (e.g. ERBB2 in renal cancer).

[EXAM WATCH] Allolactose vs IPTG: allolactose is the natural inducer (made from lactose by beta-gal). IPTG is a non-hydrolysable analog used in recombinant systems.

Lecture 5 - Slide-Specific Facts & 'If you see -> think'

Lac repressor off	Allolactose (or IPTG) binds LacI -> shape change.
Repressed Trp operon	Tryptophan binds Trp repressor.
RO for HER2: breast vs gastric	~2% vs ~10%.
HIF-1alpha in hypoxia	Stabilised -> HREs -> VEGF.
Alternative splicing in cancer	Neoantigens; drug resistance (skip drug-binding domain); pro- to anti-apoptotic switch.
YAP nuclear effect	Binds TEAD -> CTGF / JAG1 / NOTCH2 -> proliferation + EMT.
miRNA binding target	3' UTR seed sequence (~6-8 nt).

Lecture 5 - Quick Self-Test

Cover the answer column. Say it out loud. Then reveal.

Q1. When is the Lac operon maximally ON?	Lactose present + glucose LOW (LacI off + CAP-cAMP on).
Q2. What does Tamoxifen do in endometrium vs breast?	Antagonist in breast (anti-tumour). Partial agonist in endometrium - small increased cancer risk.
Q3. Order of cytokines in the NF-kB response.	TNF-alpha -> IL-1 beta -> IFN-gamma -> IL-2 -> IL-15.
Q4. Why are miRNAs great serum biomarkers?	Highly stable in circulation (exosomes / Ago-bound), unlike naked mRNA.
Q5. synNotch AND-gate logic in CAR-T?	Engineered to require two antigens before firing - reduces off-target toxicity in solid tumours.

6. Cross-Lecture Cheat Sheet

Highest-yield concepts, confusion-pairs, scenario archetypes.

> **[INFO] Diversity** = genetic variation, antibody variability, alternative splicing, meiosis, tumour heterogeneity. **Stability** = DNA repair, mitochondrial inheritance, epigenetic memory, biotech checkpoints.

High-Confusion Pairs (easy to mix up on scenario questions):

BER vs NER	BER = single damaged base; NER = bulky multi-nucleotide lesion (UV dimer).
MMR vs HR	MMR = mismatches / small indels; HR = high-fidelity DSB repair.
HR vs NHEJ	HR uses template, S/G2 only. NHEJ no template, all phases, error-prone.
cis vs trans	cis = same DNA molecule; trans = diffusible TF from elsewhere.
IgG1 vs IgG4	IgG1 -> ADCC; IgG4 -> no ADCC, longer half-life.
Autologous vs Allogeneic CAR-T	Patient-derived vs donor-derived.
Driver vs Passenger mutation	Functional cause vs bystander.
Eu- vs Heterochromatin	Active vs silent.
DBD vs LBD	DNA-binding vs ligand-binding domain of nuclear receptor.
de novo vs maintenance methylation	New mark vs copy on daughter strand.
ADCC vs CDC	Antibody-dependent cellular cytotoxicity vs complement-dependent.
Predictive vs Prognostic biomarker	Predicts *response to therapy* vs predicts *disease outcome*.

10 Most-Likely Exam Scenarios:

Genetic counsellor interpretation	Given a pedigree + BRCA test, calculate proband risk; explain why mutation does not equal certainty (gene-expression control).
Kaplan-Meier curve reading	Same target, opposite prognosis depending on cancer type. (K-Ras pan-cancer example.)
Drug-target match	Given patient's K-Ras status, recommend or exclude cetuximab; explain mechanism.
ASO / siRNA / CRISPR tool choice	Transient knockdown vs stable knockout vs precise knock-in.
MSI-H vs MSS CRC	Recommend immunotherapy or not; explain CEA's role as a surrogate marker.
Clinical trial design	Phase 1A vs 1B vs 2 vs 3; surrogate vs clinical endpoint; ~$200k per patient.
CAR-T vs CAR-NK vs CAR-M	Match to solid-tumour vs haematological indication.
Tissue-specific pharmacology	SERM antagonist here (breast), agonist there (endometrium) - tamoxifen.
Receptor occupancy	Same HER2 drug, different RO thresholds in breast vs gastric cancer.
Predictive vs prognostic biomarker	Distinguish and apply correctly to a clinical scenario.

Lecture Metadata

L1 - Introduction to Biomolecules	12 Sep 2026	~2h 24m
L2 - Basic Cell Biology	19 Sep 2026	~2h 20m
L3 - DNA and Chromatin Structure	3 Oct 2026	~2h 6m
L4 - DNA Replication, Repair, Recombination	10 Oct 2026	~2h 18m
L5 - Control of Gene Expression	17 Oct 2026	~2h 11m

> **[CLINICAL]** Good luck on the midterm - answer in English, defend your reasoning, not your facts.

15 exam-style scenario questions · drawn from Lectures 1–5

Format mirrors the midterm short-question section: a clinical/genetic vignette + data + "what do you recommend, and why"

How to use these

1. Read the vignette and the data **only** — write your answer before looking.
2. Compare against the model answer, but the marks are for **reasoning**, not for matching wording.
3. Check the **Common trap** line — that is where most marks are lost.
4. Re-do any case you got wrong after 24 hours.

Every number in these cases comes from the lectures or the slides, so the reasoning transfers directly to the exam.

Case 1 — The resistant colon cancer patient

Vignette

A 58-year-old man presents with metastatic colorectal cancer. His tumour sequencing report shows:

- **KRAS G12D mutation — detected**
- **BRAF wild-type**
- **MSI-low** (no instability)

He asks whether he should be offered **cetuximab** (an anti-EGFR monoclonal antibody). He has read online that EGFR-targeted drugs "work well in colon cancer".

Question (a) Should he receive cetuximab? Justify using the signalling pathway.

Question (b) Name one molecular feature that *would* predict cetuximab benefit.

Question (c) His tumour is MSI-low, not MSI-high. Does immunotherapy apply?

MODEL ANSWER

(a) No. EGFR and RAS sit in the same linear pathway:

```
EGFR -> RAS -> RAF -> MEK -> ERK (proliferation)
PI3K -> AKT -> mTOR (survival)
```

Cetuximab blocks **EGFR**, the receptor at the *top* of the cascade. Wild-type KRAS is a **transducer** — it only relays the signal while EGFR is firing, so blocking the receptor works. **Mutant KRAS is constitutively active**: it is stuck in the GTP-bound state and fires **regardless of whether EGFR is blocked**. The drug removes the upstream trigger but the downstream signal never stops, so the tumour keeps proliferating through RAF–MEK–ERK and PI3K–AKT–mTOR. This is a **predictive biomarker** failure — the mutation tells you the drug will not work, not that the patient is sicker.

(b) KRAS wild-type status. Optionally: BRAF wild-type, or (in other settings) EGFR amplification by FISH.

(c) No, not on this basis. Immunotherapy tracks **MSI/MMR**, not KRAS:

- **MSI-high** (defective MMR) → frameshift neoantigen burden → "hot tumour" → responds to anti-PD-1/anti-PD-L1.

- **MSI-low** here means the mismatch-repair machinery is intact, so the neoantigen load is low → "cold tumour" → checkpoint blockade is unlikely to help.

Common trap: saying "KRAS mutation means cetuximab will help because it's a known mutation" — a mutation is only useful as a biomarker if it predicts response. Here it predicts *resistance*.

Case 2 — Reading a survival curve

Vignette

You find a Human Protein Atlas entry for **ERBB2 (HER2)** summarising three Kaplan–Meier plots, all at $p < 0.001$:

Cancer	Prognostic direction
Renal cancer	**Favourable** — high ERBB2 = better survival
Endometrial cancer	**Unfavourable** — high ERBB2 = worse survival
Pancreatic cancer	**Unfavourable** — high ERBB2 = worse survival

Question ERBB2 is a therapeutic target. Why would a pharma company not simply treat every ERBB2-high patient with Herceptin?

MODEL ANSWER

Because **prognostic** and **predictive** are different things:

- **Prognostic** = the marker tells you about outcome, whatever treatment you give.

- **Predictive** = the marker tells you whether a *specific treatment* will work for you.

A marker can be **prognostic but not predictive**. ERBB2 being favourable in renal cancer does not mean blocking it helps; it may simply be a correlate of a well-differentiated, less aggressive tumour. Only in settings where anti-ERBB2 therapy has a demonstrated survival benefit — breast and gastric — is ERBB2 used as a **predictive** biomarker. Elsewhere high ERBB2 is a description of prognosis and must not be read as a treatment recommendation.

> **Common trap:** treating "high expression" as automatically equal to "treat it". Ask which evidence supports *the drug in this tumour type*.

Case 3 — A biomarker discovery you are handed

Vignette

A colleague proposes a **serum Vimentin** test for early hepatocellular carcinoma. Their stratified data:

Group	n	Serum Vimentin (ng/mL), mean ± SD
Healthy controls	64	176.0 ± 61.3
HCC, tumour ≤ 2 cm	44	271.5 ± 165.9
HCC, tumour > 2 cm	44	231.3 ± 132.3

Question (a) Which group best supports "early-stage" detection, and why does that matter?

Question (b) Give one reason this may still fail as a screening test.

Question (c) Which proteomics workflow produced the initial candidate list?

MODEL ANSWER

(a) The ≤ **2 cm** group. It has the **largest separation from controls** (271.5 vs 176.0 ng/mL). A screening test is judged on how well it separates *early disease* from healthy people, not late disease — by the time a tumour is > 2 cm it is often already symptomatic, so a large signal there buys nothing. Interestingly the > 2 cm mean is **not higher** than the small-tumour mean, so the marker is not simply tracking tumour bulk.

(b) Any of: the **overlapping distributions** (SDs of 132–166 ng/mL against a 95 ng/mL mean gap mean individual patients will fall on the wrong side); the large **standard deviations**, which imply poor specificity; the need for a **cut-off ROC/AUC** before it can be used clinically; or the fact that this was measured in **already-diagnosed patients**, not a screening population — the false-positive rate in the general population is unproven.

(c) 2D-PAGE (2-DE) — first dimension **isoelectric focusing** (charge), second **SDS-PAGE** (molecular weight) — followed by excising the spot and identifying the protein by **LC-MS/MS**.

> **Common trap:** quoting the mean difference alone. Always pair it with spread/overlap, and remember a diagnostic needs sensitivity, specificity and a defined threshold.

Case 4 — The ambiguous flow plot

Vignette

A flow cytometry run on a patient's PBMCs produces a plot. Axes: **FSC vs SSC**, plus CD3 and CD4 fluorescence.

Question (a) What do FSC and SSC measure?

Question (b) A gate has been drawn around "CD3+CD4+ events". A colleague says "that's 40% helper T cells". Why is the gate essential, and what could go wrong without it?

Question (c) If the same stained sample must be recovered alive for downstream culture, which instrument is needed and why not a slide?

MODEL ANSWER

(a) FSC ∝ cell size; **SSC ∝ internal complexity / granularity** (roughness of the membrane and internal organelles).

(b) A leukocyte sample is a **mixture**. Raw acquisition includes lymphocytes, monocytes and granulocytes. The CD3/CD4 axes do not distinguish them cleanly, and monocytes and granulocytes can fall into overlapping regions and register partial fluorescence. **Gating on the lymphocyte region (FSC/SSC) first, then gating CD3+CD4+ within that population, is what makes the 40% interpretable.** Without gating, the number is contaminated by other cell types and can be badly wrong — which is exactly why the two **cyan** dyes are also specified.

(c) FACS (Fluorescence-Activated Cell Sorting) — the same optics plus electrostatic or mechanical droplet deflection to physically divert live cells into collection tubes. A slide (or a plain analyser) destroys the sample: it stains, fixes and mounts it. If you only want to *analyse*, an analyser is enough; if you need viable cells back for culture, sequencing or therapy, you need a sorter under sterile conditions.

> **Common trap:** quoting a percentage without stating the gating hierarchy. "40% of PBMCs" and "40% of lymphocytes that are CD3+CD4+" are different numbers.

Case 5 — Manufacturing a CAR-T that keeps failing

Vignette

A team engineers CAR-T cells against a HER2-positive gastric tumour. After infusion the cells expand poorly and the tumour does not respond. Investigation shows the patients had received heavy prior chemotherapy.

Question (a) Name the three CAR-T manufacturing routes.

Question (b) Why does prior chemotherapy hurt autologous CAR-T performance?

Question (c) The team proposes an antigen-independent "safety switch" built from a Notch receptor. Explain how this changes the safety profile.

MODEL ANSWER

(a) Autologous (patient's own T cells), **allogeneic** (off-the-shelf, from healthy donors or iPSC-derived lines), **in vivo** (mRNA in lipid nanoparticles delivered to the patient's own T cells in the body, so they are engineered in place).

(b) Chemotherapy depletes and **dysfunctions the patient's T-cell compartment** — *T-cell fitness* (proliferative capacity, viability, memory phenotype) is poor. Autologous CAR-T is literally made from those damaged cells, so you are dose-limiting a product built from the worst available substrate. Allogeneic cells sidestep this because they come from a healthy, undamaged donor.

(c) A native **Notch** receptor is cleaved by ligand engagement, releasing **NICD**, which enters the nucleus as a transcription factor. Rewriting this into **synNotch** puts an inducible promoter under the control of that transcription event. This builds an **AND gate**: the CAR gene is expressed **only when antigen A (tumour cell) and antigen B (a contextual marker) are both present**.

Why it matters: a conventional CAR fires when it touches *any* cell carrying the target antigen, including healthy tissue expressing it weakly — the source of on-target/off-tumour toxicity. synNotch requires the tumour's *context* to be present as well, so normal tissue expressing antigen A alone cannot trigger the CAR. Targeting specificity rises sharply.

> **Common trap:** describing synNotch as simply "a better CAR". The key is that it changes the **logic** from OR to AND.

Case 6 — Choosing a companion diagnostic

Vignette

An ADC trial is enrolling gastric cancer patients. The protocol requires HER2 status by **FISH**, not IHC. Two patients both show HER2-positive tumour by immunohistochemistry.

Question (a) What is FISH measuring that IHC is not, and why does that matter for anti-HER2 therapy?

Question (b) One patient is HER2-positive by IHC but FISH-negative. What do you conclude about her expected benefit?

MODEL ANSWER

(a) FISH uses a **fluorescent DNA probe** complementary to the HER2 locus, and counts **gene copies per nucleus**. So it detects **amplification** — a genomic change. IHC uses an **antibody** to detect HER2 **protein**. This matters because anti-HER2 therapy in gastric cancer is licensed against **amplified** HER2; protein can be expressed at high level in a cell that has not amplified the gene, and that protein-high / copy-low situation behaves differently.

(b) She should **not** be expected to benefit. Her tumour is protein-positive without amplification. This is exactly why the protocol specifies FISH: the trial's validated biomarker is gene amplification, and IHC alone would enrol patients outside the evidence base. It also illustrates why a cheap, available assay is not automatically the right one.

> **Common trap:** saying "HER2 is positive, therefore treat". Positivity defined by the *wrong modality* is not the same indication.

Case 7 — The glioblastoma treatment decision

Vignette

Two glioblastoma patients, same age, same WHO grade, same surgical extent.

- **Patient A:** MGMT promoter **hypermethylated**
- **Patient B:** MGMT promoter **unmethylated**

Both are about to receive temozolomide plus radiotherapy.

Question (a) Whose tumour will be harmed more by temozolomide, and why?

Question (b) Which patient should have better survival, and what is this biomarker called?

Question (c) Name the two classes of methyltransferase that maintain the mark.

MODEL ANSWER

(a) Patient A — and that is *good* news, not bad. Temozolomide methylates guanine at O6. The repair enzyme **MGMT** removes that methyl group. If the **MGMT promoter is methylated, MGMT is silenced**, so the O6-methylguanine lesion persists and is converted into a lethal cross-link/kill event. **Unmethylated MGMT (Patient B)** efficiently repairs the damage, so temozolomide does much less.

(b) Patient A has the better prognosis — MGMT promoter methylation is a **predictive biomarker** for temozolomide response (and it happens to be prognostic too). The clinical shorthand is "MGMT methylation predicts benefit from alkylating chemotherapy".

(c)

- **De novo methyltransferases — DNMT3A / DNMT3B:** establish brand-new marks on previously unmethylated CpG.
- **Maintenance methyltransferase — DNMT1:** reads the hemi-methylated parent strand after replication and copies the pattern onto the new strand.

> **Common trap:** assuming "silenced gene = worse outcome". Here silencing a *repair* gene is exactly what makes the chemotherapy work.

Case 8 — A family pedigree and aneuploidy

Vignette

A couple have a child with **Down syndrome (Trisomy 21)**. Karyotyping shows 47 chromosomes. A second couple has a child with **Turner syndrome (45,X)**.

Question (a) At which meiotic stage did each error most likely occur?

Question (b) For Down syndrome, name the two possible origins.

Question (c) Why are the live-born outcomes so different?

MODEL ANSWER

(a)/(b) Both arise from **non-disjunction** — failure of chromosomes to separate — but at different divisions:

- **Trisomy 21** → gamete has **24** chromosomes, so zygote is 47.
- **Turner 45,X** → gamete has **no sex chromosome**, so zygote is 45,X.
- **Meiosis I error:** homologous chromosomes fail to separate → produces **two disomic gametes + two nullisomic** gametes → fertilisation by a normal gamete gives **trisomy** (n+1) or **monosomy** (n–1).
- **Meiosis II error:** sister chromatids fail to separate → **one disomic + one nullisomic + two normal** → also trisomy/monosomy in a different proportion.

Down syndrome typically arises from a **maternal meiosis I** error. Either parent can in principle be the source.

(c) The asymmetry is about **which chromosomes survive**. In trisomy 21 the embryo keeps 47 chromosomes, a small surplus of a small chromosome, and survives. In monosomy X the embryo is missing an entire sex chromosome —

every X-linked gene is present in only one copy — and only a fraction of such conceptions survive; most miscarry. The X chromosome is unusually gene-dense and dosage-sensitive.

> **Common trap:** saying "Turner is trisomy-like". It is a **monosomy**, and the live-birth rate differs because a whole chromosome is missing.

Case 9 — An inversion that changes expression without changing dosage

Vignette

A patient carries a large **inversion** on chromosome 7 near the *SHH* locus. Sequencing confirms the gene is intact — **copy number is unchanged**.

Question The patient has an abnormal phenotype. If no gene was gained, lost or mutated, how can expression change? What class of variant is this?

MODEL ANSWER

An inversion does not change gene **dosage** — nothing is added or removed. But it **changes genomic context**. Transcriptional regulation depends on physical proximity: an **enhancer** works by contacting a promoter, and this requires the two to be near each other in 3D chromatin. A large inversion can:

1. **Move a gene away from its enhancer**, weakening or abolishing expression; or

2. **Move a gene into contact with a different, inappropriate enhancer**, causing ectopic expression.

This is a classic example of a **structural variant that is pathogenic without being coding**, and it explains why copy-number analysis (CNV calling, FISH, karyotyping) alone would have reported this patient as normal.

> **Common trap:** concluding "no mutation, therefore no effect". The variant class to name here is a **structural variant (inversion) affecting regulatory context**.

Case 10 — A tumour that resists the obvious drug

Vignette

A patient with high-grade serous ovarian cancer has a **BRCA1** germline mutation. Her tumour responds to a PARP inhibitor for 18 months, then progresses.

Question (a) What does a PARP inhibitor do in a BRCA1-deficient cell? Explain using synthetic lethality.

Question (b) Why does resistance develop?

Question (c) What is the therapeutic-window advantage here?

MODEL ANSWER

(a) Synthetic lethality: loss of *gene A* alone is survivable; loss of *gene B* alone is survivable; losing **both** kills the cell.

- The tumour is already **A-deficient** (BRCA1 loss → defective homologous recombination).

- **HR is high-fidelity but has a backup.** Single-strand breaks are normally repaired by PARP-dependent pathways. PARP inhibition traps PARP on DNA, preventing repair of these single-strand breaks. They convert into **double-strand breaks**.

- Normal cells still have working HR and clear the damage. The BRCA1-deficient cells **cannot** — they have no HR to fall back on → cell death.

So the drug selectively kills the cells that are *already* broken in the exact pathway the drug disables.

(b) Secondary reversion / restoration of HR. Resistance typically arises through:

- **Reversion mutations** that restore BRCA1 open reading frame/function;

- **Loss of 53BP1, REV7 or Shieldin** — these normally block end resection, and removing them restores end resection and therefore HR in the BRCA1-deficient cell.

Either way the cell **regains homologous recombination** and is no longer selectively killed.

(c) Normal cells have intact HR, so they tolerate the drug at doses that kill the tumour → a **wide therapeutic window** between the effective dose and the toxic dose. This is why PARP inhibitors are far better tolerated than, say, classic cytotoxic DNA cross-linkers.

> **Common trap:** saying "PARP inhibitors damage DNA so they kill tumour cells". The selectivity comes from **synthetic lethality**, not from DNA damage per se.

Case 11 — Interpreting an MMR result

Vignette

A colorectal cancer specimen is tested:

- **MLH1, MSH2, MSH6, PMS2** — all retained by IHC
- Microsatellite panel: **stable**

Question (a) Is this tumour MSI-high? What does that imply about the immune environment?

Question (b) Which targeted therapy class becomes relevant, and by what mechanism?

Question (c) Which circulating marker would a clinician track alongside imaging, and why is a surrogate useful here?

MODEL ANSWER

(a) No — this is MSS (microsatellite stable). IHC showing all four MMR proteins retained, plus a stable microsatellite panel, means the repair machinery is intact. Consequently there are few slippage-induced **frameshifts**, so the **neoantigen load is low**: a "**cold tumour**" with little T-cell infiltration and poor response to immunotherapy.

(b) Contrast this with **MSI-high** (MMR loss → unrepaired indels at microsatellites → frameshift mutations → abundant tumour-specific neoantigens → "hot tumour" → responds to **immune checkpoint blockade**, e.g. anti-PD-1 or anti-PD-L1). Here that pathway is closed. Note also that **PD-L1** can still suppress any T cells that do infiltrate, which is why PD-1/PD-L1 blockade is tried in the MSI-H setting and generally not here.

(c) CEA (carcinoembryonic antigen). It is a fetal glycoprotein re-expressed by many colorectal tumours and is measurable in blood. It serves as a **surrogate marker**: repeated CEA gives an indication of tumour burden without the cost, radiation exposure, logistical difficulty and patient stress of serial CT/MRI. Because CEA can also be raised in benign conditions, it is tracked as a **trend**, and changes are interpreted alongside imaging rather than in isolation.

> **Common trap:** equating "MMR retained" with "no immunotherapy ever". The correct reasoning here is that the neoantigen-generating pathway is intact, so the expected benefit is low.

Case 12 — Choosing an editing strategy

Vignette

A lab must achieve one of three goals in a human cell line:

1. **Temporarily** reduce *GATA1* expression for 48 hours, then restore it.
2. **Permanently** inactivate *GATA1* for the life of the cell.
3. **Insert** a fluorescent tag at the exact start codon of *GATA1*, leaving the protein sequence unchanged.

Question For each goal, name the most appropriate tool and state which DSB-repair pathway does the work.

MODEL ANSWER

Goal	Tool	Repair pathway	Why
1. Temporary knockdown	**siRNA / shRNA**	none — RISC degrades the mRNA	You want to *reduce transcript*, not change the genome. Effect fades as the RNA is degraded and the construct dilutes.
2. Permanent inactivation	**CRISPR/Cas9**	**NHEJ**	Cas9 cuts both alleles. Error-prone NHEJ repairs the DSB with small indels; if the indel falls in the coding region it frameshifts the gene out. Because the genome is changed, the effect is inherited by every daughter cell.

3. Precise tag insertion	**CRISPR/Cas9 + donor template**	**HR** (homology-directed repair)	Supply a donor plasmid carrying the tag flanked by homology arms. The cell copies the donor into the cut site, placing the tag exactly at the start codon without altering the protein sequence.

Critical caveat: HR is **restricted to S/G2**, because a sister chromatid must exist to serve as the template. A confluent, serum-starved, G0-arrested culture — which is exactly how many cell lines are kept — will do NHEJ and give you indels instead of a clean knock-in. For goal 3 the cells must be **actively cycling and in S/G2**.

Common trap: using siRNA for a permanent knockout, or expecting a knock-in from a quiescent culture.

Case 13 — Why the topoisomerase inhibitor is in the antibody

Vignette

A new ADC, **trastuzumab-deruxtecan (T-DXd)**, is approved for HER2-positive breast cancer. It delivers **deruxtecan (DXd)**, a **topoisomerase I inhibitor**, linked to an anti-HER2 antibody.

Question (a) What does topoisomerase I normally do, and why is inhibition lethal?

Question (b) Why attach such a payload to an antibody at all, instead of giving free DXd?

Question (c) Several tumours relapse after responding well. Name one mechanism.

MODEL ANSWER

(a) During replication, helicase unwinds the helix, and **topoisomerase I** relieves the torsional strain that builds ahead of the fork by temporarily nicking one strand and re-ligating it. **Camptothecins and their derivatives trap the enzyme in the covalent cleavage complex**: the nicked DNA cannot be re-ligated, so each replication attempt converts a transient single-strand nick into an irreversible **double-strand break**, which the cell cannot repair → death. It is a replication-dependent poison: it does little to non-dividing cells, which is part of why the therapeutic window exists.

(b) The antibody is a **targeting device**: it delivers a very potent payload selectively to HER2-expressing cells, sparing normal tissue. Free DXd would be distributed everywhere and would be limited by systemic toxicity. The strategy converts a drug that is far too toxic to use alone into one that can be dosed.

(c) Any of:

- **HER2 amplification ceases / antigen is lost** — no target, so the payload never arrives.
- **Topoisomerase I mutations or reduced expression** — the target of the payload is gone.
- **HER2/HER3 or HER2/HER4 heterodimer switching** — anti-HER2 binding that alters the dimer partner.
- **Efflux pumps (e.g. ABCB1)** exporting the payload.
- **Single-strand-break repair upregulation**.

Common trap: confusing topoisomerase I with topoisomerase II — the POIs used in ADCs (DXd, SN-38) target **I**; etoposide/anthracyclines target **II**.

Case 14 — A compound that switches itself off in the right cells

Vignette

A team is engineering CAR-T cells so they activate **only inside the tumour core**. The tumour core is hypoxic; normal tissue is not. They place the CAR transgene under the control of a **Hypoxia Response Element (HRE)** promoter.

Question (a) Explain why an HRE-controlled CAR stays quiet in normal tissue.

Question (b) The construct fails and CAR-T kill normal cells. What single upstream step did they most likely get wrong?

Question (c) How could they add a second, independent safety layer, and what logic would it give?

MODEL ANSWER

(a) The mechanism runs through **HIF-1alpha**.

- **Normoxia:** HIF-1alpha is continuously produced but immediately **degraded** (via prolyl hydroxylation → VHL recognition → ubiquitin–proteasome). It never accumulates, so it cannot activate the HRE. The CAR gene is **off**.

- **Hypoxia:** the hydroxylases cannot work, so HIF-1alpha **stabilises**, dimerises with HIF-1beta, enters the nucleus and binds the **HRE**.

- Therefore the CAR protein is produced **only where oxygen is low** — i.e. in the tumour core. The hypoxia response element is being used as a **molecular oxygen sensor**.

(b) The most likely failure is a **constitutive HIF-1alpha** problem — e.g. HIF-1alpha not properly degradable, or an HRE-driven promoter with a **leaky/basal activity**, or a **VHL-pathway defect** in the T cells. If HIF-1alpha is not degraded in normoxia, the gate is open everywhere and the CAR is expressed in all cells. The CAR then fires on any normal cell expressing the antigen — classic **on-target, off-tumour toxicity**.

(c) Add a **synNotch AND gate**: put the HRE-driven HIF response *upstream of a synNotch element* rather than driving the CAR directly.

- synNotch requires antigen **A** (tumour cell) and antigen **B** (a contextual marker) before the CAR gene is transcribed.

- Combined with the hypoxia input, the CAR is expressed only when **(tumour cell present) AND (tumour context present) AND (hypoxic)** — three inputs instead of one. Any single failure is survivable.

> **Common trap:** using one environmental cue alone as the sole safety gate. Single-gate designs are fragile; the lecturer's point is that you stack independent orthogonal controls.

Case 15 — The isoform that predicts drug resistance

Vignette

In liver cancer, sequencing finds a **single-nucleotide substitution** that creates a **novel splice site**. The resulting protein lacks part of a domain and is secreted rather than membrane-bound. Patients whose tumours express it show significantly **worse overall survival**; mice overexpressing it show larger, more invasive tumours.

Question (a) What mechanism produced this change — one gene, but how many proteins?

Question (b) Name two clinical consequences of this event.

Question (c) If you designed an ASO to block it, what would it silence, and how is that different from blocking with an anti-sense DNA gap-mer?

MODEL ANSWER

(a) Alternative splicing. The spliceosome chooses splice sites from the pre-mRNA. A single base substitution can:

- create a **new splice acceptor/donor site**, or

- abolish a normal site,

so the exon is included, skipped or partially included in a different combination. One gene therefore yields a different **isoform** — and here a truncated, **soluble** version that no longer has the transmembrane domain, so it leaks out of the cell.

(b) Two valid consequences:

1. **A secreted biomarker.** Because the isoform lacks the membrane anchor, it is measurable in **serum** — turning a tissue event into a non-invasive liquid-biopsy test for prognosis.

2. **A neoantigen / immune escape route.** The novel isoform-derived peptides are tumour-specific. They can generate neoantigens that alter immunogenicity — helping convert a "cold tumour" to a "hot" one or vice versa — and, in the drug context, removing or truncating the **drug-binding domain** produces **treatment resistance**.

(c) An **ASO (antisense oligonucleotide)** is a short single-stranded DNA/RNA analogue that binds the complementary RNA and recruits RNase H, degrading the target mRNA. It therefore silences **both** the normal and the aberrant isoform — it acts pre-mRNA/mRNA.

An **anti-sense DNA gap-mer** binds a specific region (usually across the splice junction or an exon–intron boundary) and **blocks the spliceosome**, so only the intended isoform is suppressed. It is more selective — it spares the normal protein, which matters when you do not want to lose the wild-type function.

> **Common trap:** describing alternative splicing as a mutation that damages the gene. The gene sequence changed by one base, but the mechanism that produces many proteins is **splicing**, and the therapeutic handle is the **splice site**,

not the coding change.

Case 16 — Reading the lac operon logic correctly

Vignette

An expression construct is engineered using the bacterial lac system for *E. coli* expression of a human protein. The operator is disrupted in one clone and intact in another. IPTG is used to induce.

Question (a) What is IPTG, and why is it used instead of lactose in an industrial setting?

Question (b) In the operator-disrupted clone, transcription is constitutive. Give one production advantage and one production problem.

Question (c) Why is *E. coli* a poor host for a human protein that requires glycosylation, and what is used instead?

MODEL ANSWER

(a) IPTG (isopropyl β-D-1-thiogalactopyranoside) is a **non-metabolisable lactose analogue**. It binds and inactivates the **LacI** repressor, but the cell cannot break it down, so induction is **sustained and reproducible** rather than depending on how much lactose happened to be consumed. In minimal or poorly-reducing media, lactose would not induce reliably at all.

(b)

- **Advantage:** no induction step is needed — expression is automatic, which simplifies the process.

- **Problem:** you cannot *tune* expression. Constitutive expression is usually undesirable because it burns the host's resources and may produce toxic intermediate levels.

A better design is an **IPTG-inducible construct**: a *lac operator* (or *lacN*-controlled repressor) placed downstream of the promoter so IPTG gives dose-dependent control — a **tunable** induction rather than an on/off switch.

(c)

- *E. coli* is a **prokaryote** — it has no ER, no Golgi and **no glycosylation machinery**, so it cannot attach the carbohydrate trees that many human proteins need to fold and function.

- Hence **mammalian CHO cells** (Chinese hamster ovary) are used for complex, glycosylated human proteins; **codon optimisation** adapts the gene sequence to the host's codon usage.

- A further reason to delay induction: if expression starts before exponential growth, the burden and misfolding drive **inclusion bodies**.

Common trap: treating the lac operon as three independent inputs. It is a **two-input** gate: **LacI (lactose)** AND **CAP–cAMP (glucose)** both have to permit transcription.

Use this when a case is close:

Marks	What earns them
1 mark	Correct drug / technique named.
2 marks	Correct mechanism stated (which pathway, which protein, why).
3 marks	Correct mechanism **and** the clinical consequence — the "so what for the patient".
4 marks	All of the above **plus** naming the biomarker type (predictive / prognostic / surrogate).

The lecturer's own words: *"multiple choice is for you to pass the exam, but if you want to get a full mark or a better result, you'd better do perform well for the short questions, which emphasize on data analysis and also the scenario questions."*

Every case above is a **scenario question with data**. Answer in **English**.

9. Active-Recall Flashcards (120 Q/A)

Cover the answer column. Say it out loud. Reveal. Mark what you hesitated on; revisit in 24 hours.

Section A - Lecture 1 (Q1-Q22)

Q1. What are the four classes of biomolecules?	Carbohydrates, lipids, proteins, nucleic acids.
Q2. Monosaccharides, disaccharides, polysaccharides - give one example of each.	Mono: glucose (or fructose). Di: sucrose. Poly: glycogen (animal), starch (plant), cellulose (plant).
Q3. What is the Warburg effect?	Cancer cells preferentially convert pyruvate to lactate even when oxygen is present (aerobic glycolysis). Frees intermediates for biosynthesis.
Q4. Which metabolite in the Warburg pathway is most often measured as a biomarker?	Lactate.
Q5. Two volatile-organic-compound biomarkers used for non-invasive cancer detection?	VOCs in exhaled breath - for lung and gastric cancer.
Q6. Endotoxin that triggers septic shock via TLR4 - give the lipid component that does the binding.	Lipopolysaccharide (LPS); its **lipid A** moiety binds **TLR4**.
Q7. State the four levels of protein structure.	Primary (sequence) -> Secondary (alpha-helix / beta-sheet) -> Tertiary (3D fold) -> Quaternary (subunit assembly).
Q8. Define a protein "domain".	A structurally and functionally independent region that can fold into its native shape even if excised from the rest of the polypeptide.
Q9. Modern technique for sequencing proteins; legacy method.	Modern: **LC-MS/MS**. Legacy: **Edman degradation**.
Q10. Common chemical liability in therapeutic proteins: Asn -> Asp at high pH.	**Deamidation** - can change drug binding/efficacy.
Q11. Three components of a typical antibody-drug conjugate (ADC)?	Monoclonal antibody (targeting) + linker + cytotoxic payload (e.g., topoisomerase inhibitor like DXd/SN-38).
Q12. What does a "dual-payload ADC" try to overcome?	Single-drug resistance - by co-delivering two agents (e.g., topo + DDR inhibitor).
Q13. Define the Hippo pathway cascade and what it does to YAP.	Cell-cell contact -> MST1 -> LATS1 phosphorylate **YAP** -> YAP excluded from the nucleus -> no proliferation.
Q14. Phenotype of *MST1* double-knockout?	YAP stays unphosphorylated -> enters nucleus -> uncontrolled organ overgrowth -> cancer.
Q15. Kaplan-Meier lesson: a single tyrosine-kinase target can have what opposite prognostic effects?	High expression -> poor survival in pancreatic cancer but **better** survival in renal cell carcinoma.

Q16. 2D-PAGE separates proteins by two dimensions - what are they?	**Isoelectric focusing (IEF)** by charge, then **SDS-PAGE** by molecular weight.
Q17. Soluble serum biomarker found by the lecturer's group in early HCC via 2D-PAGE.	**Vimentin** - detectable in serum by ELISA.
Q18. Griffith's + Avery-MacLeod-McCarty proved what?	That **DNA** (not protein) is the genetic material - DNase destroyed the transforming principle; proteases did not.

Q19. What is codon optimisation and why is it needed in CHO expression?	Replacing synonymous codons with host-preferred codons to maximise translational efficiency in CHO cells (codon-usage bias).
Q20. K-Ras: why does mutant K-Ras predict cetuximab failure?	Cetuximab blocks upstream EGFR; mutant K-Ras is constitutively active downstream -> EGFR blockade is bypassed.
Q21. Frameshift = insertion/deletion not in multiple of three. Three downstream consequences?	(i) Premature stop -> truncated protein; (ii) completely novel amino-acid sequence; (iii) potentially tumour-specific **neoantigens** for immunotherapy.
Q22. Why are miRNAs preferred as serum biomarkers over mRNAs?	miRNAs are far more stable in circulation.

Section B - Lecture 2 (Q23-Q46)

Q23. In flow cytometry, what do FSC and SSC measure?	**FSC** = cell size; **SSC** = granularity / internal complexity.
Q24. CD3+CD4+ cells are which T-cell subset?	Helper T cells.
Q25. What does FACS add to flow cytometry?	Physical sorting of viable cells under sterile conditions for downstream culture/assay.
Q26. Three techniques for isolating a pure cell population from heterogeneous tissue.	**FACS**, **Laser Capture Microdissection (LCM)**, **gradient centrifugation** (or specific antibody-conjugated magnetic beads).
Q27. FISH vs IHC - what does each detect?	FISH = **DNA probes** -> gene copy number (e.g., HER2 amplification). IHC = **antibodies** -> protein expression.
Q28. Why is confocal microscopy needed for subcellular localisation?	It rejects out-of-focus light via a pinhole -> optical sectioning + 3D reconstruction.
Q29. Three reasons mitochondria are likely endosymbiotic in origin.	(i) Double membrane; (ii) own circular DNA; (iii) bacterial-like ribosomes.
Q30. Why is mtDNA inherited maternally only?	Paternal mitochondria are tagged (ubiquitin) and destroyed after fertilisation.
Q31. Define the mitochondrial "threshold effect" in disease.	Clinical severity depends on whether the mutant/wild-type mtDNA ratio exceeds a tissue-specific tolerance.
Q32. Cell-cycle phases in order.	G1 -> S -> G2 -> M (+ cytokinesis) -> G0 (quiescence).
Q33. What is the role of cyclin-CDK complexes?	Drive cell-cycle transitions (e.g., CDK4/6 + cyclin D -> G1 -> S).
Q34. Two ways to measure/synchronise cell-cycle position in vitro.	**PI staining + flow cytometry** (DNA content); **serum starvation** (drop FBS to ~0.5-1%) -> G1 arrest.
Q35. Binucleated/multinucleated cells after drug treatment suggest the drug blocks what?	**Cytokinesis** (the contractile ring) - nuclear division proceeded but cytoplasmic division did not.
Q36. Meiosis I separates _____ ; Meiosis II separates _____ .	Meiosis I = **homologous chromosomes**; Meiosis II = **sister chromatids**.
Q37. Two apoptotic pathways (extrinsic + intrinsic) converge on which enzymes?	**Caspases** (caspase cascade) + mitochondrial cytochrome c release.
Q38. How is autophagy exploited in cancer?	Tumours in hypoxic cores use autophagy to recycle nutrients and survive stress.
Q39. Name the four Yamanaka factors.	**Oct3/4, Sox2, Klf4, c-Myc** (OSKM).
Q40. Gold-standard in vivo test for pluripotency after iPSC reprogramming?	**Teratoma assay** - inject into immunodeficient mouse -> tissues from all three germ layers (ectoderm, mesoderm, endoderm).

Q41. CRISPR vs siRNA/shRNA - which gives permanent vs transient gene knockdown?	**CRISPR** = permanent (genome-level cut). **siRNA/shRNA** = transient (post-transcriptional).
Q42. IgG1 vs IgG4 therapeutic antibody - which triggers ADCC?	**IgG1** (via Fc-FcR on NK cells). IgG4 has poor Fc binding -> no ADCC but longer half-life.
Q43. Three flavours of CAR-T therapy - autologous, allogeneic, in vivo - what each means.	**Autologous** = patient-derived; **allogeneic** = off-the-shelf from healthy donor or iPSC; **in vivo** = LNP-mRNA delivered directly into patient.
Q44. Two markers used to isolate CTCs from blood and why both.	**EpCAM+** (positive selection of epithelial tumour cells) + **CD45-** (negative selection to exclude white blood cells).
Q45. Why might a CTC's biomarker profile differ from the primary tumour?	**EMT (Epithelial-Mesenchymal Transition)** alters surface marker expression.
Q46. MSIs vs MSS tumours - which responds to anti-PD-1/PD-L1?	**MSI-H** (high neoantigen load = "hot tumour") responds.

Section C - Lecture 3 (Q47-Q63)

Q47. G-C has how many hydrogen bonds; A-T has how many?	G-C = 3; A-T = 2.
Q48. Genome-wide resolution limit of G-banding karyotyping?	~0.5-10 Mb - large structural changes only. Point mutations need sequencing.
Q49. Euchromatin vs heterochromatin - which is transcriptionally active?	**Euchromatin** = active (relaxed); heterochromatin = silent (condensed).
Q50. Mechanism: how do nucleosomes move along DNA?	(i) intrinsic DNA-histone affinity, (ii) **ATP-dependent chromatin remodelling complexes**.
Q51. Two ends of a chromosome - short arm vs long arm - names.	**P-arm** (short, "petit"); **q-arm** (long).
Q52. Hayflick limit - what's happening?	Telomeres shorten each division; at threshold -> senescence -> no further division.
Q53. Telomerase - what kind of enzyme, with what template?	A **reverse transcriptase** with an internal **RNA template**.
Q54. De novo vs maintenance methyltransferase - which does what?	**De novo** = establishes new methylation marks; **maintenance** = copies existing marks onto daughter strands after replication.
Q55. What does CIMP stand for, and which cancer type features it?	**CpG Island Methylator Phenotype** - particularly in **colorectal cancer**, with poorer prognosis.
Q56. Two non-invasive methods for detecting tumour methylation markers clinically.	**Stool DNA tests** (e.g., Cologuard) and **plasma cell-free DNA (cfDNA)** methylation assays.
Q57. Bisulfite sequencing - what chemical conversion is key?	Sodium bisulfite converts **unmethylated cytosine to uracil** (read as T after PCR); methylated C is preserved.
Q58. Histone code nomenclature - what does H3K9me3 mean?	Histone H3, lysine at position 9, **tri-methylation**.
Q59. What does an HDAC inhibitor do, and how does it create "gene addiction"?	Restores acetylation -> reactivates silenced tumour suppressors; also induces **PD-L1** expression -> tumour becomes dependent on PD-L1 -> vulnerable to anti-PD-1/PD-L1.
Q60. Aneuploidy -> non-disjunction happens in meiosis I vs meiosis II - what results?	MI failure -> homologues don't separate; MII failure -> sister chromatids don't separate. Both -> gametes with wrong chromosome count.
Q61. CNV mechanism in meiosis.	**Non-Allelic Homologous Recombination (NAHR)** - unequal crossover at repetitive sequences.
Q62. Driver vs passenger - how to distinguish them with bioinformatic integration?	Cross-reference **CNV profiles + mRNA expression** - true drivers show concordant copy gain + transcript upregulation (e.g., *NPM1*).
Q63. Textbook dietary CNV example.	**AMY1** (amylase) - higher copy number in high-starch-eating populations.

Section D - Lecture 4 (Q64-Q83)

Q64. Meselson-Stahl proved what?	**Semi-conservative** DNA replication.
Q65. Four core enzymes of replication and what each does.	**Helicase** (unwind); **topoisomerase** (relieve torque); **primase** (RNA primer); **DNA polymerase** (extend + proofreading).
Q66. Okazaki fragments form on which strand and why?	**Lagging strand** - because DNA can only be synthesised 5'->3' and the template runs 3'->5' relative to the fork; synthesis is therefore discontinuous.
Q67. Role of DNA ligase.	Seals nicks between Okazaki fragments by forming phosphodiester bonds.
Q68. Why do eukaryotes need thousands of replication origins?	Eukaryotic chromosomes are very long; a single origin couldn't replicate them within S phase. Multiple origins -> multiple bidirectional **bubbles**.
Q69. Telomerase reactivation in HCC - clinical implication?	Re-lengthened telomeres = **independent prognostic marker** for higher recurrence and worse survival.
Q70. 3'->5' exonuclease activity of polymerase: name and clinical cost.	**Proofreading** - but can also excise nucleoside-analogue drugs (drives antiviral / anticancer resistance).
Q71. Two p53-related functions relevant to DNA integrity.	Cell-cycle arrest on damage; mismatch detection/proofreading assist; trigger apoptosis if unrepairable.
Q72. Three exogenous sources of DNA damage.	**UV light** (pyrimidine dimers); **ionising radiation** (DSBs); **viral integration** (e.g., HBV DNA into host genome).
Q73. BER vs NER - what does each fix?	**BER** = single damaged base; **NER** = bulky multi-nucleotide lesion (e.g., UV dimer).
Q74. Microsatellite instability mechanism.	Defective **MMR** -> unrepaired slippage at di/tri-nucleotide repeats -> insertions/deletions -> frameshift mutations.
Q75. Why is MSI-H a "hot tumour"?	Many frameshift mutations -> many neoantigens -> strong T-cell recognition -> responsive to checkpoint inhibitors.
Q76. CEA - what is it and how is it used clinically?	Embryonic glycoprotein re-expressed by some cancers; **circulating surrogate biomarker** for tumour burden (esp. CRC) - used in lieu of repeat CT/MRI.
Q77. Approximate per-patient cost of a US clinical trial.	~**US$200,000** per patient.
Q78. HR vs NHEJ - template? cell-cycle phase?	**HR** = sister-chromatid template; **S/G2 only**. **NHEJ** = no template; works **all phases** but error-prone.

Q79. Define synthetic lethality.	Loss of gene A alone is tolerable; loss of gene B alone is tolerable; loss of *both* is lethal - exploitable when cancers lack gene A (e.g., BRCA).
Q80. Why does PARP inhibition kill BRCA1/2 tumours selectively?	BRCA- tumour can't do HR -> relies on PARP-mediated single-strand break repair -> PARP inhibition traps the cell in unrepairable damage.
Q81. V(D)J recombination - what enzyme class does it use?	**NHEJ** machinery.

Q82. SCID - link to NHEJ.	NHEJ defects -> no functional V(D)J recombination -> no mature T/B cells -> severe combined immunodeficiency.
Q83. CRISPR outcome depends on the DSB repair pathway - what does each pathway produce?	**NHEJ** = small indels -> gene **knock-out**; **HR** (with donor template) = precise **knock-in** / gene editing.

Section E - Lecture 5 (Q84-Q110)

Q84. Cis-acting elements include promoter, enhancer, silencer. What defines "cis"?	Acts on the **same DNA molecule** on which it sits.
Q85. A trans-acting factor is usually a _____.	A **transcription factor** (or diffusible regulatory protein) encoded elsewhere.
Q86. Lac operon: how is it activated and why does low glucose matter?	Lactose -> allolactose binds LacI (repressor) -> repressor off. Low glucose -> ^cAMP -> CAP-cAMP binds promoter -> activates Pol II recruitment.
Q87. Trp operon - constitutive vs repressible?	**Repressible** - tryptophan binds Trp repressor -> shuts down its own biosynthesis.
Q88. IPTG - what is it and why is it used in recombinant expression?	Non-metabolised lactose analogue that binds LacI -> induces expression without being used by cells.
Q89. Why delay IPTG induction until exponential phase?	Avoids toxic pre-expression that causes **misfolding / inclusion-body aggregation**.
Q90. Exon vs intron.	**Exon** = coding, retained in mature mRNA; **intron** = non-coding, spliced out.
Q91. Three consequences of aberrant alternative splicing in cancer.	(i) Novel **neoantigens**; (ii) **Drug resistance** by excluding binding domains; (iii) Switch between isoforms (e.g., pro-apoptotic -> anti-apoptotic).
Q92. Standard reporter gene for promoter mapping.	**Luciferase** (with serial deletion constructs of the candidate promoter).
Q93. Three functional domains of a nuclear receptor.	**NTD** (regulatory), **DBD** (DNA binding), **LBD** (ligand binding).
Q94. What is HER2 doing that Herceptin blocks?	HER2 heterodimerises with other HER family members -> drives proliferation. Herceptin blocks that dimerisation.
Q95. Tamoxifen tissue-selective action - what tissue + effect.	**Antagonist in breast** (anti-tumour); **partial agonist in endometrium** (small ^ endometrial cancer risk).
Q96. Receptor occupancy (RO) for HER2: breast vs gastric cancer.	~**2%** RO required in breast cancer; ~**10%** RO required in gastric cancer.
Q97. NF-kappaB pathway: where is it held and how is it released?	Held in cytoplasm by **IkappaB**; on LPS/TNF signal -> IkappaB degraded -> NF-kappaB translocates to nucleus.
Q98. Order of cytokines in the NF-kappaB response.	**TNF-alpha -> IL-1beta -> IFN-gamma -> IL-2 -> IL-15**.
Q99. What is the "amplification loop" in NF-kappaB signalling?	Secreted pro-inflammatory cytokines act back on the cell -> further NF-kappaB activation -> rapid escalation of defence.
Q100. HIF-1alpha - what stabilises it and what does it induce?	**Hypoxia** stabilises HIF-1alpha -> translocates to nucleus -> binds **HREs** -> induces **VEGF** and other adaptive genes.

Q101. HRE-controlled CAR-T - what is the synthetic-biology logic?	CAR expression only in hypoxic tumour microenvironment -> spares healthy normoxic tissues.

Q102. Hippo-YAP - what happens when the Hippo pathway is lost?	YAP stays unphosphorylated -> accumulates in the nucleus -> drives proliferation, organ overgrowth, and tumour formation.
Q103. Notch mechanism in one sentence.	Juxtacrine signal: ligand binding triggers proteolytic cleavage of Notch -> NICD enters nucleus as a transcription factor.
Q104. synNotch - what's the "AND-gate" logic?	CAR-T only fires when *both* antigen A (tumour-specific) AND antigen B (contextual) are present -> much higher specificity.
Q105. lncRNA - length threshold and function.	>200 nt, non-coding; scaffold/guide for chromatin modifiers, TFs, and proteins.
Q106. Three mRNA structural elements that control stability/translation.	**5' cap**, **poly-A tail length**, **3' UTR**.
Q107. How does a miRNA silence a target mRNA?	Binds complementary **seed sequence** in 3' UTR (imperfect match) -> mRNA degradation and/or translational repression.
Q108. Why are miRNAs especially good serum biomarkers?	Highly stable in circulation (encapsulated in exosomes or bound to Argonaute proteins) - unlike naked mRNA.
Q109. How does an mRNA vaccine work?	Exogenous mRNA delivered (e.g., LNP) -> translated by host cell -> processed -> peptides presented on **MHC** -> activate T- and B-cells.
Q110. Two alternatives to CAR-T for solid tumours and their rationale.	**CAR-NK** and **CAR-M (macrophages)** - better tissue infiltration and different effector mechanisms.

Section F - Exam Scenario Rapid-Fire (Q111-Q120)

Q111. A patient with metastatic colorectal cancer shows a KRAS mutation. Should you give cetuximab?	No - KRAS mutation predicts **cetuximab resistance** (downstream constitutive activation).
Q112. A glioblastoma biopsy is MGMT-promoter-methylated. Drug of choice?	**Alkylating agent (e.g., temozolomide)** + radiation - MGMT silencing means less DNA repair of the alkylation damage.
Q113. A patient's tumour is MSI-H. Best therapy class?	**Immune checkpoint inhibitors** (anti-PD-1 / anti-PD-L1) - high neoantigen load + hot tumour.
Q114. Patient is BRCA1+ ovarian cancer. Which targeted drug class?	**PARP inhibitor** (synthetic lethality with HR deficiency).
Q115. Two patients both have HER2+ tumours but in different tissues. Should you dose Herceptin the same?	No - **receptor occupancy** differs by tissue (~2% in breast vs ~10% in gastric cancer).
Q116. You need a permanent gene knockout in a cell line. What tool?	**CRISPR/Cas9** with NHEJ-repaired DSB -> small indel -> frameshift.
Q117. You want to test which promoter regions drive expression of gene X. Method?	**Luciferase reporter assay** with serial deletions of the promoter region.
Q118. A patient on tamoxifen has thickened endometrium - why?	Tamoxifen is a **SERM** - antagonist in breast but **partial agonist in endometrium** -> small risk of endometrial proliferation/cancer.
Q119. Which two pathways together generate antibody diversity?	**Somatic V(D)J recombination** (uses NHEJ) + **somatic hypermutation** (affinity maturation).
Q120. A student says "patient has a point mutation so we need G-banding to find it." Push back: why?	G-banding resolution is ~0.5-10 Mb - **cannot detect point mutations**; need PCR/Sanger or NGS.